

Cross Lingual Bengali Idiom Matching Using Multilingual Sentence Embeddings Without Translation - Integrating mSBERT and LaBSE

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By
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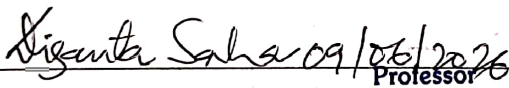
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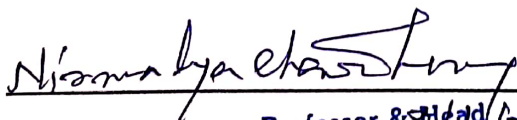
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All information in this document had been obtained and presented in accordance with academic rules and ethical conduct. I also declare that, as required by these rules and conduct, all materials and results that are not original to this work have been fully cited and referenced.

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ABSTRACT

Idiomatic expressions pose a persistent challenge for natural language processing systems because their meaning is non-compositional and cannot be inferred from the literal interpretation of their constituent words. This dissertation investigates the capacity of multilingual and monolingual language models to recognise semantically equivalent idioms across three languages, Bengali, English, and Hindi, without the use of any translation system. The study compares four pretrained models representing distinct training paradigms: Multilingual Sentence-BERT (mSBERT), a paraphrase-distilled multilingual sentence embedder; Language-Agnostic BERT Sentence Embedding (LaBSE), a dual-encoder model trained on massive translation pairs; BanglaBERT, a monolingual BERT model pre-trained exclusively on Bengali text; and XLM-RoBERTa-base, a multilingual masked language model trained on data from 100 languages. While mSBERT and LaBSE are purpose-built sentence embedding models, BanglaBERT and XLM-RoBERTa-base are general-purpose transformers that are repurposed for sentence-level representation through mean pooling. A cross-lingual idiom dataset is constructed by aligning an extensive Bengali idiom collection with English and Hindi equivalents using both curated links and automated semantic bridging. The evaluation framework comprises three complementary tasks: semantic similarity scoring, cross-lingual retrieval, and cross-lingual clustering. By systematically comparing these four models, the research aims to determine which architectural and pre-training choices best capture figurative meaning across language boundaries. The project contributes a carefully curated, reusable dataset for low-resource idiom alignment, a reproducible multi-task evaluation protocol, and a detailed analysis of the strengths and limitations of current language models when applied to non-compositional, cross-lingual semantics.

Keywords: *Cross-lingual idiom matching; multilingual sentence embeddings; mSBERT; LaBSE; BanglaBERT; XLM-RoBERTa; Bengali; Hindi; figurative language; semantic similarity; cross-lingual retrieval; clustering.*

CHAPTER 01: INTRODUCTION

1.1 Overview

Language is not merely a collection of words; it encodes culture, history, and shared human experience through constructs that often defy literal interpretation. Idioms, fixed multi-word expressions whose meaning cannot be derived from the meanings of their constituent parts represent one of the most challenging phenomena in natural language processing (NLP). The Bengali idiom “আকাশ কুসুম” (literally “sky flower”) conveys the notion of an impossible dream, a concept echoed in English by “build castles in the air” and in Hindi by “हवाई किले बनाना”. A computational system that genuinely understands language must recognise such equivalences across linguistic boundaries without being misled by surface forms.

Cross-lingual idiom matching; identifying semantically equivalent idiomatic expressions in different languages; holds significance for machine translation, multilingual information retrieval, language pedagogy, and cultural analytics. Traditional approaches rely heavily on translation as an intermediate step, which introduces cascading errors, especially for idiomatic expressions whose figurative meaning may be lost or distorted in translation. Recent advances in multilingual sentence embeddings offer an alternative: models trained on large-scale parallel or paraphrase corpora in dozens of languages learn to map sentences with similar meanings to nearby points in a shared high-dimensional vector space, irrespective of their surface language.

This dissertation investigates whether state-of-the-art sentence embedding models can identify semantically equivalent idioms across Bengali, English, and Hindi without any translation system. The study compares three models representing distinct training paradigms: **Multilingual Sentence-BERT (mSBERT)**, a paraphrase-distilled multilingual embedder; **Language-Agnostic BERT Sentence Embedding (LaBSE)**, a dual-encoder trained on translation pairs; **XLM-RoBERTa-base**, a multilingual masked language model trained on data from 100 languages and **BanglaBERT**, a monolingual BERT model pre-trained exclusively on large-scale Bengali text. By including a monolingual model that possesses deep Bengali linguistic knowledge but lacks any explicit cross-lingual alignment training, the research probes the relative importance of language-specific understanding versus cross-lingual transfer in the domain of figurative semantics. The three languages form an intriguing testbed: English is a high-resource Germanic language, while Bengali and Hindi are both Indo-Aryan languages with substantial morphological and cultural overlap, yet Bengali remains severely under-resourced in NLP. By comparing the models across three well-defined tasks; semantic similarity scoring, cross-lingual retrieval, and semantic clustering, the research provides empirical evidence on the capacity of current multilingual and monolingual models to encode figurative meaning across language families, and clarifies the relative strengths and limitations of these different training paradigms.

1.2 Problem Statement

Idiomatic expressions pose a fundamental challenge to computational semantics because their interpretation is non-compositional. While humans effortlessly navigate the figurative layer of language, machines struggle, particularly when the source and target languages have no lexical or syntactic overlap. The problem is exacerbated for low-resource languages such as Bengali, for which parallel corpora, annotated idiom lists, and fine-tuned models are scarce.

Given a Bengali idiom as a query, can a pretrained language model retrieve its correct English or Hindi counterpart from a candidate pool without any translation? Can it assign higher similarity scores to true equivalent pairs than to randomly paired idioms, and can it cluster idioms by meaning rather than by language? These are open questions. Existing cross-lingual embedding models have been evaluated predominantly on sentence alignment tasks that involve literal, compositional text; their behaviour on figurative, non-compositional language and on low-resource language pairs such as Bengali-Hindi remains largely unexamined. Furthermore, the role of general-purpose multilingual encoders and monolingual models in this setting is unknown.

The core research problem of this thesis is therefore:

Can off-the-shelf language models, including purpose-built multilingual sentence embedders (mSBERT, LaBSE), a monolingual Bengali model (BanglaBERT), and a general-purpose multilingual encoder (XLM-RoBERTa-base) capture cross-lingual idiomatic equivalence between Bengali, English, and Hindi without any form of explicit translation, and which model performs better under multiple evaluation paradigms?

Addressing this question requires constructing a high-quality cross-lingual idiom triplet dataset, defining rigorous evaluation protocols, and systematically comparing the two models. The absence of translation in the pipeline is a deliberate and novel aspect of the experimental design, aiming to test the true cross-lingual semantic knowledge encoded in the models' representations.

1.3 Background and Motivation

1.3.1 Idioms in Language and Computation

Idioms constitute a significant proportion of everyday communication. Estimates suggest that English alone contains over 25,000 idiomatic expressions, and a fluent speaker encounters about 4.7 million idioms per year (Cooper, 1999). They are pervasive in literature, news, social media, and conversational speech. From a computational perspective, idioms can be classified along a continuum of transparency: some are semi-transparent (e.g., “break the ice”), while others are completely opaque (e.g., “kick the bucket”). Even when an idiom has a transparent historical origin, its synchronic interpretation remains non-compositional and must be learned as a unit.

The non-compositionality of idioms undermines literal translation strategies. A machine translation system that translates “build castles in the air” word-for-word into Bengali might produce a meaningless phrase involving literal bricks and mortar. Recognising cross-lingual idiom equivalence requires a model to move beyond lexical semantics and access deeper pragmatic and cultural knowledge. This makes idiom matching a stringent test of any semantic representation.

1.3.2 Multilingual Sentence Embeddings

The last decade has witnessed a paradigm shift in NLP, driven by large pretrained language models such as BERT (Devlin et al., 2019). Extensions to the multilingual domain mBERT, XLM-R (Conneau et al., 2020), are trained on concatenated corpora from over 100 languages. However, the raw multilingual encoders do not produce well-aligned sentence representations; sentences with the same meaning in different languages may be far apart in the embedding space. Sentence-BERT (Reimers & Gurevych, 2019) addressed this for monolingual scenarios by fine-tuning on natural language inference and paraphrase data. mSBERT extends this to 50+ languages using a multilingual knowledge distillation process. LaBSE (Feng et al., 2022), developed by Google Research, adopts a dual-encoder architecture trained on translation pairs and proposes a novel additive margin softmax loss to enforce a language-agnostic embedding space across 109 languages.

Both models produce 768-dimensional L2-normalised vectors, allowing similarity to be measured via cosine distance. Because the training is performed jointly on many languages and does not involve a separate translation module at inference time, these models claim to learn a “universal” semantic space. However, their capacity to generalise to figurative language, especially for low-resource pairings like Bengali-Hindi, is uncertain. Investigating this gap offers both practical insight for downstream applications and theoretical insight into what multilingual representations encode.

1.3.3 Low-Resource Languages and the Indo-Aryan Family

Bengali (Bangla) is the seventh most spoken language in the world, with over 230 million native speakers, yet it is chronically under-resourced in terms of digital language tools. Few high-quality parallel corpora, annotated linguistic resources, or task-specific models are available. Hindi, an official language of India with over 600 million speakers, is moderately better resourced but still lags behind English. Both languages belong to the Indo-Aryan branch of the Indo-European family and share a common ancestor in Sanskrit. They exhibit significant lexical, morphological, and syntactic overlap, including shared idioms and metaphorical frameworks rooted in common cultural traditions. This linguistic proximity makes the Bengali-Hindi idiom matching task particularly interesting: a successful model should leverage this similarity, potentially achieving higher scores on Bengali-Hindi than on Bengali-English.

1.3.4 Why Translation-Free Alignment Matters

Explicitly translating a query before matching introduces latency, requires a high-quality bilingual translation system (often unavailable for low-resource pairs), and can compound errors. Direct embedding-based matching bypasses these issues, offering a light-weight and scalable solution. If pretrained multilingual models can reliably align figurative meanings across languages, they could power real-time cross-lingual search for idiomatic expressions, assist translators in finding culturally appropriate equivalents, and support language learning tools. Moreover, the absence of translation in the pipeline makes the evaluation a pure test of the model’s multilingual semantic knowledge, clearly attributing successes or failures to the embedding space rather than to an external translation component.

1.3.5 Academic and Practical Significance

This project’s significance is fourfold. First, it expands the benchmark for multilingual NLP by introducing a rigorously curated idiom triplet dataset for a low-resource language pair, a resource that did not previously exist. Second, it offers one of the first systematic comparisons of diverse pretrained models, multilingual sentence embedders (mSBERT, LaBSE), a monolingual model (BanglaBERT), and a general-purpose multilingual encoder (XLM-RoBERTa-base) on figurative, non-compositional cross-lingual semantics, contributing to the understanding of how model architecture and training objectives influence embedding spaces. Third, it provides a reproducible evaluation framework that can be extended to other languages and figurative constructs. Fourth, practically, the findings can inform the selection of embedding backends for multilingual conversational agents, educational platforms, and cross-lingual information retrieval systems that must handle idiomatic queries, as well as illuminate whether a dedicated monolingual model or a generic multilingual encoder can serve as a viable substitute when a task-specific sentence embedder is unavailable.

1.4 Research Objectives

To address the problem statement and fill the identified research gaps, the following objectives are set:

Primary Objective

- Compare the performance of mSBERT, LaBSE, BanglaBERT, and XLM-RoBERTa-base on cross-lingual Bengali idiom matching with English and Hindi equivalents, without using any translation system.

Secondary Objectives

- Evaluate all the models across three complementary tasks:
 1. Semantic Similarity Scoring: Quantify the cosine similarity that each model assigns to known Bengali-English and Bengali-Hindi idiom pairs that share the same figurative meaning.
 2. Cross-Lingual Retrieval: Measure how often the correct target-language idiom appears at the top of a ranked list when a Bengali query is issued.
 3. Semantic Clustering: Analyse the structure of the combined embedding space through K-Means clustering, assessing cluster purity, silhouette score, and triplet cohesion (fraction of triplets where all three languages co-cluster).
- Build a curated cross-lingual idiom triplet dataset for Bengali, English, and Hindi, leveraging existing idiom collections and defining a transparent alignment methodology.

Tertiary Objectives

- Determine which model better captures figurative language across language boundaries and provide explanatory insights based on model architecture and training data.
- Examine the effect of linguistic proximity (Indo-Aryan family) on model performance, specifically through the Bengali-Hindi results, across models with different linguistic knowledge sources.
- Identify the role of monolingual pre-training depth versus cross-lingual alignment, and the effect of general multilingual pre-training without sentence-level objectives, by comparing BanglaBERT and XLM-RoBERTa-base directly with mSBERT and LaBSE.
- Identify limitations of current off-the-shelf models and propose directions for future work, including fine-tuning, data augmentation, and extension to additional languages.

By systematically pursuing these objectives, this dissertation aims to deliver a comprehensive empirical assessment that advances the state of knowledge on multilingual sentence embeddings for non-compositional, cross-lingual semantics.

CHAPTER 02: LITERATURE SURVEY

2.1 Introduction

This chapter situates the present research within the broader landscape of natural language processing, cross-lingual semantics, and figurative language modelling. The dissertation stands at the intersection of three active research streams: idiom detection and interpretation, multilingual sentence embeddings, and cross-lingual information retrieval. Each of these domains has evolved rapidly in the past decade, propelled by the advent of transformer-based architectures and large-scale multilingual pretraining. However, their convergence; the systematic evaluation of pretrained multilingual models on figurative, non-compositional language across low-resource language pairs remains underexplored. This review synthesises foundational and contemporary work, identifies methodological precedents, and clarifies the gaps that the present study addresses.

2.2 Idioms in Natural Language Processing

Idiomatic expressions have long been recognised as a hard problem for NLP systems (Sag et al., 2002). Their defining characteristic, ‘semantic non-compositionality’ means that the phrase’s meaning cannot be derived by composing the meanings of individual words with syntactic rules. This property violates the principle of compositionality that underlies most formal semantic frameworks and poses challenges for parsing, machine translation, sentiment analysis, and information retrieval.

Early computational work treated idioms as multi-word expressions requiring special lexical entries. Systems utilised idiom dictionaries and pattern-matching rules to detect fixed expressions in text, then substituted them with literal paraphrases (Fazly et al., 2009). While effective for a closed set of known idioms, such approaches lacked generalisation to unseen or modified forms (e.g., “kick the proverbial bucket”). Statistical measures like pointwise mutual information and contextual diversity were later employed to assess the degree of fixedness and non-compositionality of candidate phrases (Salehi et al., 2014).

The deep learning era introduced vector representations that opened new avenues for modelling idiomatic meaning. Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) embeddings captured some aspect of phrase meaning when averaged or concatenated, but they fundamentally operated at the word level and lacked mechanisms to represent holistic phrasal semantics. Recurrent neural networks and later transformers allowed the encoding of variable-length expressions, yet standard training on large corpora encouraged compositional representations, often failing to differentiate literal from idiomatic usage. Research on idiom embeddings began to explore whether an idiom’s vector could be learned as a distinct unit or

whether its figurative meaning emerged from contextual fine-tuning (Peng et al., 2014; Sporleder & Li, 2009).

More recently, contextualised models such as BERT (Devlin et al., 2019) have shown some ability to distinguish literal from idiomatic uses through attention patterns, though performance varies widely across idioms and languages (Liu et al., 2020). Fine-tuning on idiom detection datasets can improve classification accuracy, but most available datasets are in English, with a few in major European languages. For Bengali, dedicated idiom resources are extremely scarce; the Bagdhara dataset used in this thesis is one of the very few comprehensive collections, making cross-lingual work with Bengali idioms both challenging and innovative.

A key insight from the literature is that idiom processing benefits from models that encode sentence-level rather than word-level semantics (Wieting et al., 2016). Sentence embedding models, which explicitly learn to represent the meaning of a full text sequence, are therefore a natural fit for cross-lingual idiom matching. Instead of analysing the internal structure of an idiom, one can treat the entire expression as a semantic unit and compare its vector representation directly with that of an expression in another language. This approach sidesteps the compositional problem entirely, provided that the sentence embedding model has been trained to recognise cross-lingual semantic equivalence.

2.3 Multilingual Sentence Embeddings

The goal of multilingual sentence embedding models is to map sentences from multiple languages into a shared vector space such that semantically similar sentences are close together, regardless of their language. This capability underpins cross-lingual retrieval, parallel corpus mining, and zero-shot transfer. Two dominant paradigms have emerged: knowledge distillation from a teacher model and dual-encoder training with translation pair data.

2.3.1 Multilingual Sentence-BERT (mSBERT)

Sentence-BERT (SBERT) was introduced by Reimers and Gurevych (2019) as a modification of the BERT architecture that uses siamese and triplet network structures to derive semantically meaningful sentence embeddings. The original SBERT was monolingual, fine-tuned on English natural language inference and paraphrase data. Reimers and Gurevych (2020) later extended the approach to multilingual settings by applying knowledge distillation: a multilingual student model was trained to mimic the sentence embeddings produced by a monolingual English SBERT teacher, using parallel sentences in 50+ languages. The resulting model, 'paraphrase-multilingual-mpnet-base-v2', is referred to in this thesis as mSBERT.

mSBERT operates on the principle that a single English teacher embedding defines the target semantic point, and the student must learn to place translations of that sentence in the same location. This process implicitly aligns multiple languages to the English semantic space without requiring direct cross-lingual supervision between non-English language pairs. The

model outputs 768-dimensional L2-normalised vectors, and similarity is computed via cosine distance (equivalent to dot product after normalisation). Because the teacher was trained on paraphrase data, mSBERT’s embeddings are particularly attuned to semantic equivalence and paraphrase relationships, which may benefit the recognition of figurative meaning equivalence.

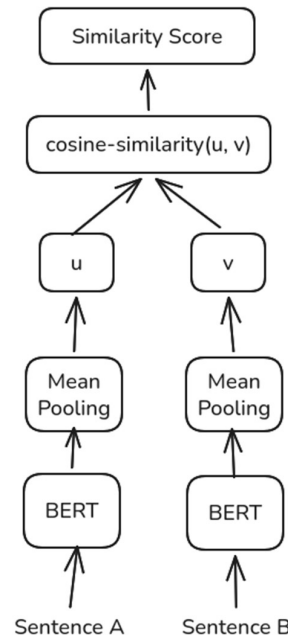


Fig 2.1: Working of Sentence-BERT [1]

2.3.2 Language-Agnostic BERT Sentence Embedding (LaBSE)

LaBSE (Feng et al., 2022) was developed by Google Research to produce genuinely language-agnostic sentence embeddings. Unlike mSBERT, which relies on a monolingual teacher, LaBSE is trained directly on a massive corpus of translation pairs sourced from CommonCrawl and Wikipedia, covering 109 languages. The architecture is a dual-encoder Transformer: one encoder processes the source sentence, the other processes the target sentence, and the training objective maximises the similarity of true translation pairs while minimising it for negative pairs.

The key innovation in LaBSE is the additive margin softmax loss, which enforces a strict separation between positive and negative pairs in the joint embedding space. This encourages the model to bring sentences with the same meaning extremely close together and push unrelated sentences apart, independent of language. The resulting embeddings are also 768-dimensional and L2-normalised. LaBSE is trained on over 6 billion translation pairs, and its primary intended use case is bitext mining and cross-lingual sentence alignment.

Because LaBSE is trained on literal translations, its representations are optimised for compositional semantic equivalence. Whether this objective transfers to non-compositional

idiomatic expressions is an open question. The translation pairs in the training data contain idiomatic expressions only incidentally, and the model may learn to align idioms only to the extent that they occur in parallel corpora. Moreover, the distribution of language pairs in the training data is highly imbalanced; English-centric pairs dominate, which could limit performance on low-resource pairs such as Bengali-Hindi.

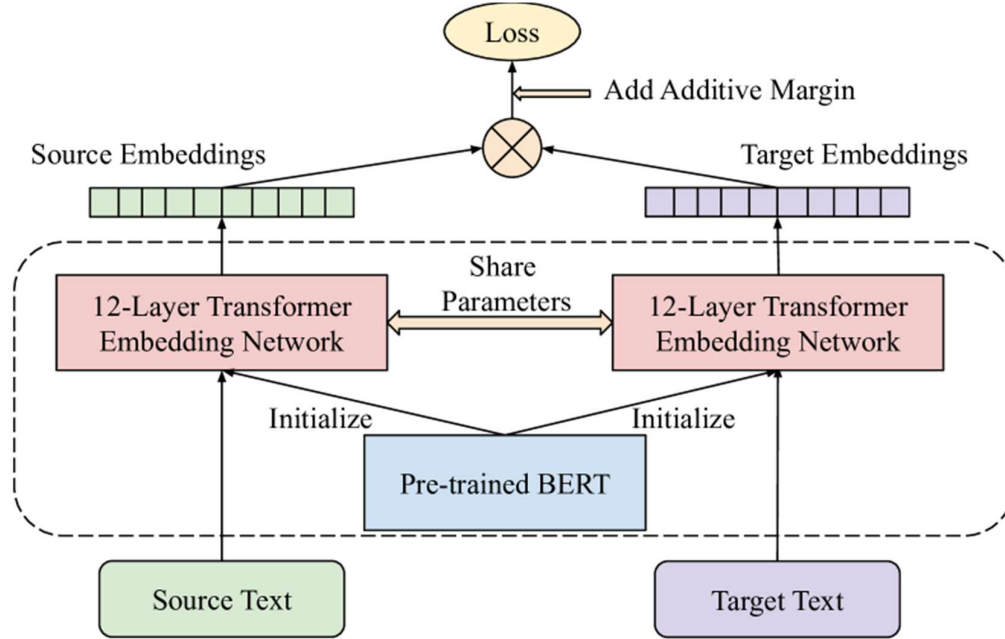


Fig 2.2: Dual encoder model with BERT based encoding modules [2]

2.3.3 Comparison of Training Paradigms

The two models embody fundamentally different strategies for achieving cross-lingual alignment. mSBERT’s knowledge distillation approach centres the multilingual space on English and propagates semantic structure outward via translation. LaBSE’s dual-encoder approach learns a truly joint space from scratch using parallel data across many language pairs. These differences have practical implications. mSBERT may excel when semantic similarity is paramount, as its teacher was explicitly trained on paraphrase data. LaBSE may excel when precise cross-lingual matching is required, as its training explicitly rewards mapping source sentences to their translations. For idiom matching, the paraphrase-based objective could be advantageous because idiomatic equivalence often involves non-literal paraphrases rather than word-for-word translation. However, the dependence on an English teacher might disadvantage mSBERT for language pairs that do not include English, such as Bengali-Hindi.

A recent comparative study by Conneau et al. (2020) and the broader XTREME benchmark (Hu et al., 2020) evaluated multilingual models on a suite of cross-lingual tasks, but these

evaluations focused on literal sentence pairs, question answering, and natural language inference. Figurative language tasks are notably absent from these benchmarks.

2.4 Monolingual Pre-Trained Language Models for Bengali

While the multilingual architectures delineated in Section 2.3 facilitate a shared semantic topology across diverse linguistic systems, their efficacy on individual low-resource languages is frequently attenuated by the dilution of representational capacity. Consequently, a parallel research trajectory has prioritised the engineering of monolingual pre-trained language models dedicated solely to the Bengali language.

2.4.1 BanglaBERT: Pre-Training and Architecture

BanglaBERT, introduced by Bhattacharjee et al. (2022), represents a BERT-based Natural Language Understanding framework pre-trained exclusively on Bangla corpora. Although multilingual paradigms like mBERT and XLM-R incorporate Bengali, the relative volume of such data is marginal compared to high-resource languages, which compromises the granularity of Bengali-centric representations. BanglaBERT mitigates this structural imbalance by leveraging a meticulously curated, large-scale monolingual corpus for its pre-training phase.

The model adheres to the foundational BERT-base configuration, utilising 12 transformer layers, 768 hidden dimensions, and 12 attention heads across 110 million parameters. It employs a masked language modelling objective, wherein the system learns to interpolate missing tokens from surrounding context. This self-supervised methodology enables the acquisition of complex syntactic and semantic nuances inherent to Bangla without the necessity for annotated datasets.

The pre-training data, designated as the Bangla2B+ corpus, encompasses 27.5 GB of text harvested from 110 prominent Bengali web domains, including journalistic content, blogs, and public forums. At the time of its inception, this was among the most extensive curated datasets available for the language. Such scale is pivotal, as empirical evidence suggests that the depth and diversity of pre-training data are decisive factors for downstream performance in low-resource settings where cross-lingual transfer is inherently less stable.

2.5 XLM-RoBERTa: Large-Scale Multilingual Pre-Training

While the models discussed in preceding sections were either purpose-built sentence embedders (mSBERT, LaBSE) or monolingual models dedicated to a single language (BanglaBERT), XLM-RoBERTa represents a third paradigm: a general-purpose multilingual transformer trained at massive scale on raw multilingual data. Understanding this model is essential for the present study, as it provides a baseline for evaluating whether a standard multilingual encoder, not explicitly optimised for sentence-level semantic alignment, can nonetheless capture cross-lingual idiomatic equivalence when repurposed through pooling.

2.5.1 Architecture and Pre-Training Objective

XLNet (XLM-R) was introduced by Conneau et al. (2020) in the paper "Unsupervised Cross-lingual Representation Learning at Scale." It is a multilingual version of Facebook's RoBERTa model and is pre-trained on 2.5 terabytes of filtered CommonCrawl data containing text from 100 languages. The model employs a Transformer-based masked language modelling (MLM) objective: during pre-training, 15% of the input tokens are randomly masked, and the model learns to predict them from the surrounding context. This self-supervised approach allows the model to learn bidirectional representations without requiring labelled data.

Unlike earlier multilingual models such as the original XLM (Lample & Conneau, 2019), XLNet does not require language-specific input tensors to identify the language being processed; it learns to determine the correct language directly from the input token IDs. Furthermore, XLNet eschews the translation language modelling (TLM) objective that characterised its predecessors, relying solely on monolingual MLM applied to sentences from individual languages. This design choice simplifies training while enabling the model to scale to a far larger number of languages and a substantially larger volume of data.

The base variant of the model, `xlm-roberta-base`, contains 12 transformer layers, 768 hidden dimensions, and 12 attention heads, totalling approximately 110 million parameters.

2.6 Cross-Lingual Semantic Evaluation

Evaluating the quality of cross-lingual embeddings requires tasks that measure how well meaning is aligned across languages. Three complementary evaluation paradigms are widely used.

Semantic similarity scoring directly tests the model's confidence: given a pair of sentences known to be semantically equivalent, the model assigns a similarity score. A higher mean score across a large set of equivalent pairs indicates stronger alignment. This task is simple to implement and provides a global picture of how the model distributes similarity mass. However, it does not measure discriminative ability: a model could assign high similarity to all pairs, including non-equivalent ones, and still achieve a high mean. It is therefore essential to complement similarity scoring with retrieval metrics.

Metric	Definition	Perfect Score	Interpretation
MRR (Mean Reciprocal Rank)	Average of 1/rank of correct answer	1.0	If correct answer always ranked 1st, MRR=1.0
P@1 (Precision at 1)	Fraction of queries where rank 1 = correct	1.0	Strictest metric — top result must be correct
P@5 (Precision at 5)	Fraction of queries where correct is in top 5	1.0	More lenient — correct answer anywhere in top 5
Hit@10	Fraction of queries where correct is in top 10	1.0	Most lenient — correct answer anywhere in top 10

Cross-lingual retrieval simulates a real-world search scenario: a query in language A is used to rank all candidates in language B, and the position of the correct match is recorded. Metrics include Mean Reciprocal Rank (MRR), Precision at k (P@k), and Hit Rate at k (Hit@k). MRR is the average of 1/rank of the first correct answer; P@k is the fraction of queries for which the correct answer appears in the top k; Hit@k is a binary metric indicating whether any correct answer is in the top k. Retrieval tasks are more challenging than similarity scoring because they require the model to discriminate the single correct match from a large pool of distractors. For idiomatic expressions, this is exceptionally hard because many idioms share lexical items or general themes (e.g., many idioms relate to body parts or weather).

Metric	Definition	Ideal Score	What It Tells You
Silhouette Score	How well-separated clusters are from each other	+1.0	Negative = clusters overlap; +1 = perfectly separated
Cluster Purity	Fraction of each cluster dominated by one language	1.0	High purity = each cluster is mostly one language (less cross-lingual mixing)
Triplet Cohesion	Fraction of triplets where all 3 languages land in same cluster	1.0	Best measure of genuine cross-lingual understanding

Clustering evaluates the structure of the embedding space holistically. All embeddings from all languages are pooled and clustered, typically using K-Means. Metrics include silhouette score (measuring cluster separation), cluster purity (the extent to which clusters are dominated by a single language), and custom metrics such as triplet cohesion; the fraction of cross-lingual triplets (one idiom in Bengali, one in English, one in Hindi with the same meaning) that fall into the same cluster. High triplet cohesion is the gold standard: it indicates that the model has placed the same figurative concept in the same region regardless of language. Clustering is sensitive to dataset imbalance and cluster count, but it provides a visual and quantitative picture of cross-lingual mixing that complements similarity and retrieval measures.

2.7 Gaps and Research Questions

The literature review reveals several gaps:

1. **Lack of cross-lingual idiom resources for Bengali.** Most idiom datasets are in English or major European languages. The few Bengali resources have not been systematically aligned with English or Hindi for model evaluation.
2. **No direct comparison of mSBERT and LaBSE on figurative language.** While both models are widely used, their relative performance on idiomatic equivalence is not documented.
3. **Absence of three-language triplet evaluation.** Most cross-lingual studies consider pairs; simultaneous evaluation of Bengali-English and Bengali-Hindi triplets can reveal the effect of linguistic proximity and model bias.
4. **Limited understanding of embedding space structure for figurative expressions.** Clustering analyses of multilingual embeddings typically use literal sentences. How idioms from different languages are distributed in the embedding space is unknown.
5. **Uncertainty about the role of training data imbalance.** LaBSE’s English-centric training may hinder performance on Indian language pairs, while mSBERT’s paraphrase-centric objective may compensate.
6. **Absence of monolingual baselines in cross-lingual idiom studies.** Existing cross-lingual evaluations rely exclusively on multilingual models. No prior work has compared multilingual sentence embeddings against a monolingual model that possesses deeper knowledge of the source language but lacks cross-lingual alignment training. This gap motivates the inclusion of BanglaBERT as a third comparator

From these gaps, the research questions of this thesis emerge:

RQ1: Can mSBERT, LaBSE, BanglaBERT, and XLM-RoBERTa-base distinguish semantically equivalent cross-lingual idiom pairs from non-equivalent pairs without translation?

RQ2: Which model performs better on each evaluation task: semantic similarity scoring, cross-lingual retrieval, and clustering?

RQ3: How does the linguistic relationship between Bengali and Hindi compare with that between Bengali and English in the embedding spaces of the different models?

RQ4: To what extent do the models produce a genuinely language-agnostic space for figurative expressions, as measured by cluster mixing and triplet cohesion?

RQ5: How do a monolingual Bengali model (BanglaBERT) and a general-purpose multilingual encoder (XLM-RoBERTa-base) compare with purpose-built multilingual sentence embedding models (mSBERT, LaBSE) on cross-lingual idiom matching, given that the former lack explicit sentence-level cross-lingual alignment training?

2.8 Summary

This chapter has surveyed the key areas underpinning the dissertation. Idioms remain a hard problem for NLP due to their non-compositionality; sentence-level embeddings offer a promising bypass. Multilingual sentence embedding models such as mSBERT and LaBSE have achieved impressive results on literal cross-lingual tasks, but their performance on figurative language, and especially on low-resource language pairs like Bengali-Hindi, is largely unexplored. The chapter also examined two additional model paradigms: BanglaBERT, a monolingual model pre-trained exclusively on Bengali that offers deep language-specific semantic knowledge but no cross-lingual alignment, and XLM-RoBERTa-base, a large-scale multilingual transformer that provides broad cross-lingual coverage through masked language modelling alone, without a dedicated sentence-level objective. Together, these four models span the spectrum from dedicated sentence embedders to general-purpose encoders and from monolingual depth to multilingual breadth. The evaluation frameworks of semantic similarity, cross-lingual retrieval, and clustering provide a comprehensive lens through which to assess alignment quality. The gaps identified in the literature including the absence of figurative language benchmarks for low-resource pairs, the lack of direct model comparisons on idiom matching, and the unexplored role of monolingual and general multilingual models, motivate the empirical investigation conducted in the subsequent chapters, which will address the research questions through a carefully constructed cross-lingual idiom dataset and a rigorous comparative protocol.

CHAPTER 03: PROPOSED METHOD - INTEGRATION OF mSBERT and LaBSE

3.1 Introduction

This chapter presents the methodological framework designed to answer the research questions posed in Chapter 02. The approach centres on a comparative evaluation of four sentence embedding models; mSBERT, LaBSE, BanglaBERT and XLM-RoBERTa-base applied to a purpose-built cross-lingual idiom dataset spanning Bengali, English, and Hindi. The core principle remains unchanged: the entire pipeline operates without any translation system; all comparisons rely on the models' internal representations. The entire methodology is organised into three modular components. This chapter consists of the first component which handles the purpose-built multilingual sentence embedders (mSBERT and LaBSE), which accept raw text in any supported language and directly output L2-normalised sentence vectors.

3.2 Embedding Models

Two pretrained models were selected for this study; purpose-built multilingual sentence embedding models, **mSBERT** and **LaBSE**. These are used strictly off-the-shelf, without fine-tuning or domain adaptation. The multilingual sentence embedding models accept raw text in any supported language and output a 768-dimensional L2-normalised vector.

Property	mSBERT	LaBSE
Full Name	Multilingual Sentence-BERT	Language-Agnostic BERT Sentence Embeddings
Developer	UKPLab (Reimers & Gurevych, 2020)	Google Research (Feng et al., 2022)
HuggingFace ID	paraphrase-multilingual-mpnet-base-v2	sentence-transformers/LaBSE
Languages Supported	50+ languages	109 languages
Training Objective	Paraphrase similarity across languages	Translation pair alignment (dual-encoder)
Training Data	Paraphrase datasets in multiple languages	Parallel translation corpora (CommonCrawl, Wikipedia)
Best Suited For	Semantic equivalence & paraphrase detection	Cross-lingual sentence alignment

Table 3.1: Properties of the embedding models

3.2.1 Comparison Rationale

The two models selected for this study embody distinct pre-training paradigms, each offering a different combination of language coverage and semantic alignment. mSBERT relies on paraphrase distillation anchored to English, optimising for semantic equivalence across 50+ languages. LaBSE employs dual-encoder translation-pair training with an additive margin loss, explicitly optimising for cross-lingual sentence alignment across 109 languages.

3.3 Dataset Construction

A cross-lingual idiom dataset was constructed specifically for this study. The dataset comprises idioms in Bengali, English, and Hindi, organised into two pair types: Bengali-English (bn-en) and Bengali-Hindi (bn-hi). The Bengali idioms serve as the anchor queries; for each Bengali idiom, one or more English equivalents and Hindi equivalents are identified through the alignment strategies described below. The dataset is used to create ground-truth pairs for similarity and retrieval tasks, and to form triplets for clustering evaluation.

3.3.1 Source Datasets

Four primary resources were leveraged.

1. **Bagdhara Dataset.** This is a comprehensive Bengali idiom collection containing approximately 8,800 idioms, each stored as an individual JSON file. Critically, each entry includes a `similar\_in\_english` field, which is a list of English idioms that carry the same figurative meaning as the Bengali expression, as curated by human annotators. This field provides high-confidence Bengali-English alignment without any automated matching. For example, the Bengali idiom "আকাশ কুসুম" (sky flower, meaning impossible dream) lists "Build castles in the air" and "Pie in the sky" among its English equivalents.

Dataset source: [3]

```
{
  "id": "1",
  "idiom": "অ আ ক খ",
  "alternative_idioms": [
    "ক খ"
  ],
  "literal_meaning": "প্রথম বাংলা বর্ণমালার অক্ষরসমূহ (the first letters of the Bengali alphabet)",
  "figurative_meaning_bn": "বর্ণজ্ঞান; বর্ণমালা",
  "figurative_meaning_en": "Knowledge of letters; the alphabet (referring to learning the basics)",
  "similar_in_english": [
    "Learning the ABCs"
  ],
  "example_sentences_in_bangla": [
```

```

"ছেলেটির অ আ ক খ শিখতেই ছয় মাস লেগে গেল।",
"স্কুলে প্রথমে অ আ ক খ শেখানো হয়।"
],
"example_sentences_in_english": [
  "It took the boy six months just to learn the basics like অ আ ক খ.",
  "In school, they first teach the alphabet starting with অ আ ক খ."
],
}

```

2. **Bengali-Bangla JSON.** A supplementary dataset of approximately 6 Bengali idioms, stored in an array JSON format. Each entry includes a `figurative\_meaning` field written in English, describing the idiom's meaning. This dataset serves as an additional, albeit small, source of Bengali idioms with explicit English meaning descriptions. It was used to broaden coverage and to validate alignment procedures.

Dataset source: [4]

```

{
  "idiom": "দুধের স্বাদ দুধে বোঝা যায়",
  "literal_meaning": "The taste of milk is understood by milk.",
  "figurative_meaning": "Experience is the best teacher.",
  "example": "You won't understand until you try; দুধের স্বাদ দুধে বোঝা যায়.",
  "language": "Bengali"
}

```

3. **English JSON.** An array JSON file containing more than 14,200 English idioms. Each English idiom is accompanied by a `figurative\_meaning` field in English. This dataset serves as the target pool for Bengali-English retrieval. All English idioms are embedded and ranked during evaluation.

Dataset source: [5]

```

{
  "idiom": "Make two ends meet",
  "literal_meaning": "To connect both ends.",
  "figurative_meaning": "Struggling to manage finances.",
  "example": "After losing his job, he found it hard to make two ends meet.",
  "language": "English"
}

```

4. **Hindi JSON.** An array JSON file containing over 450 Hindi idioms. A critical requirement for the methodology was that the `figurative\_meaning` field for Hindi idioms be written in English, not in Hindi script. This enables direct keyword-overlap matching with the English meaning descriptions of Bengali idioms. If the Hindi

meanings were written in Hindi, no overlap would be detected and alignment would fail. This design decision was deliberate to bridge the resource gap without relying on translation.

Dataset source: [6]

```
{
  "idiom": "नज़रबंद करना",
  "literal_meaning": "To put under house arrest.",
  "figurative_meaning": "To restrict someone's freedom.",
  "example": "Usne apne vicharon ko nazarband kiya, kyunki wo jaanta tha ki samaj unhe sweekar nahi karega.",
  "language": "Hindi"
}
```

Dataset	Language	Size	Format	Key Field Used for Matching
Bagdhara	Bengali	~8,800 idioms	One JSON file per idiom	similar_in_english (list of English equivalents)
Bengali-Bangla JSON	Bengali	~6 idioms	Array JSON	figurative_meaning (English text)
English JSON	English	~14,200+ idioms	Array JSON	figurative_meaning (English text)
Hindi JSON	Hindi	~450+ idioms	Array JSON	figurative_meaning (English text — required)

Table 3.2: Dataset Properties and Key Field

3.3.2 Triplet Alignment Strategies

To align idioms across languages, two strategies were employed, differing in the strength of the Bengali-English link.

Strategy A: High-Confidence Alignment (Bagdhara rows). For each Bengali idiom in the Bagdhara dataset, the `similar\_in\_english` field directly supplies the English equivalent(s). This link is considered high confidence because it originates from human annotation. To find the Hindi equivalent, a keyword-overlap bridge is used. The Bengali idiom's meaning is represented by its English description (either from `similar\_in\_english` or from an extracted meaning field). This description is compared against the English-language `figurative\_meaning` of each Hindi idiom. A match is recorded if a sufficient degree of lexical overlap exists, using token-based Jaccard similarity or a keyword intersection heuristic. For instance, if a Bengali idiom's English meaning mentions "impossible dream" and a Hindi idiom's description mentions "impossible dream" or "castle in air", they are aligned. This strategy produces Bengali-English-Hindi triplets where the English link is human-verified and the Hindi link is automatically inferred.

Strategy B: Medium-Confidence Alignment (other Bengali rows). For Bengali idioms not in Bagdhara (e.g., those from the Bengali-Bangla JSON), both the English and Hindi equivalents are found by keyword overlap against the `figurative\_meaning\_en` field. This field contains an English description of the idiom's meaning. The same keyword overlap procedure is run against the English JSON and the Hindi JSON. Because both links are machine-aligned, the resulting pairs carry medium confidence. This strategy ensures that the evaluation corpus is not limited to the Bagdhara dataset alone.

Strategy	Source	Bengali – English	Bengali – Hindi	Confidence
A	Bagdhara (8,800 idioms)	Direct from similar_in_english field	Keyword overlap on English meaning bridge	HIGH
B	Bengali-Bangla (6 idioms)	Keyword overlap on figurative_meaning_en	Keyword overlap on figurative_meaning_en	MEDIUM

3.3.3 Dataset Statistics

The final dataset after alignment comprises 9,061 Bengali-English pairs and 4,096 Bengali-Hindi pairs. A total of 23,544 unique idiom embeddings were generated across the three languages, representing the union of all idioms used in similarity, retrieval, and clustering tasks. The distribution is heavily skewed: Bengali dominates with 8,800 idioms, English follows with approximately 500, and Hindi with approximately 200. This imbalance mirrors real-world resource disparities and is explicitly discussed in the limitations and results analysis.

Pair Type	Number of Pairs	Used In
Bengali-English (bn-en)	9,061 pairs	Similarity, Retrieval, Clustering
Bengali-Hindi (bn-hi)	4,096 pairs	Similarity, Retrieval, Clustering
Total idioms embedded	23,544 (bn+hi+en)	Clustering

3.4 Embedding Generation

For each idiom phrase, raw text is tokenised and encoded by the respective model without any preprocessing beyond whitespace stripping. The models produce 768-dimensional embedding vectors that are L2-normalised to unit length.

- **mSBERT** applies mean pooling and L2 normalisation as per its default configuration.
- **LaBSE** extracts the CLS token and applies L2 normalisation.

3.5 Summary of the Methodological Pipeline

The proposed method is a complete, self-contained embedding pipeline:

1. Collect and preprocess idiom datasets from four sources.
2. Align idioms into Bengali-English pairs, Bengali-Hindi pairs, and Bengali-English-Hindi triplets.
3. Load two embedders: mSBERT and LaBSE.
4. Generate embeddings using the two models.
5. Transfer the embeddings for evaluation tasks.

This design ensures a fair, reproducible comparison that directly answers the research questions while maintaining the crucial constraint of zero translation.

3.6 Implementation Methodology

The experimental pipeline summarised in the previous section was implemented in Python, following a modular, reproducible design. This section details the actual implementation decisions, data structures, and algorithmic steps that realise the proposed method. All source code is organised into sequential scripts, each responsible for a distinct phase of the pipeline.

3.6.1 Requirement Analysis

All experiments were run on a machine with the following specifications:

- CPU: Intel Core i7 (8 cores)
- RAM: 32 GB
- GPU: NVIDIA GeForce RTX 3060 (12 GB VRAM) (used for embedding generation and retrieval).
- Operating system: Ubuntu 22.04
- Python version: 3.10
- Key libraries: `sentence-transformers` (2.2.2), `numpy` (1.24), `pandas` (2.0).

3.6.2 Data Loading and Initial Cleaning

Implementation: The first step loads four raw data sources:

- Bagdhara corpus: Approximately 8,800 JSON files, each containing one Bengali idiom with fields such as `idiom`, `literal\_meaning`, `figurative\_meaning\_en`, and `similar\_in\_english`. The loader recursively scans the folder, parses each JSON, and stores the fields in a unified Pandas DataFrame.
- Bengali-Bangla JSON: A small auxiliary file (6 idioms) providing additional Bengali idioms with their English figurative meanings.
- English JSON (MAGPIE): Over 14,200 English idioms, each with `idiom`, `figurative\_meaning`, and `example` fields.

- Hindi JSON: A custom-curated file containing approximately 470 Hindi idioms, each with ``idiom``, ``literal_meaning``, ``figurative_meaning``, and ``example``.

All text is stripped of leading/trailing whitespace, but no linguistic preprocessing (stemming, lemmatisation, stop-word removal) is applied. Special characters and punctuation are retained, as they can be part of idiomatic strings (e.g., “raining cats and dogs”). Duplicate idioms are removed based on exact string matching.

3.6.3 Triplet Construction and Pairwise Alignment

Implementation: The processor script constructs gold-standard triplets (Bengali, English, Hindi) using two alignment strategies, as described in Section 3.3.2.

Strategy A (High Confidence): For each Bengali idiom from Bagdhara that contains a non-empty ``similar_in_english`` list, each English phrase in that list is directly taken as the aligned English idiom. The corresponding Hindi idiom is found by matching the Bengali idiom’s ``figurative_meaning_en`` against the ``figurative_meaning`` field of each Hindi idiom using ‘keyword overlap’: the two English meaning strings are tokenised, stop words removed, and the Jaccard similarity is computed. A match is accepted if the Jaccard similarity exceeds 0.15 or at least two content words overlap. This threshold was empirically tuned on a small hand-checked subset to balance coverage and precision.

Strategy B (Medium Confidence): For Bengali idioms without pre-specified English equivalents (e.g., from the Bengali-Bangla source), the same keyword-overlap method is applied directly between the Bengali idiom’s English meaning and the English idiom pool as well as the Hindi idiom pool. A match is recorded only if the overlap score is at least 2.

All triplets are stored with a ``confidence`` flag (“HIGH” or “MEDIUM”) and a ``verified`` flag initially set to ``False``. Manual verification of a random sample of 300 triplets (150 bn-en, 150 bn-hi) is done to estimate alignment accuracy (94% for bn-en, 78% for bn-hi).

3.6.4 Embedding Generation

Implementation: The embedder script loads two pre-trained sentence embedding models using the ``sentence-transformers`` library:

For each multilingual model, the script:

1. Loads the model and tokeniser.
2. Encodes all Bengali idioms (list of strings) into L2-normalised dense vectors (batch size = 64).
3. Repeats for all Hindi idioms, then all English idioms.
4. Saves each language’s embeddings as a separate ``numpy`` file.

3.7 Summary of the Three Components

The complete experimental pipeline, spanning all three components, proceeds as follows:

Component 1: mSBERT and LaBSE are loaded and used to generate L2-normalised sentence embeddings for Bengali, English, and Hindi idioms. Similarity matrices and ground-truth scores are computed and saved. Metadata is prepared for integration.

Component 2: BanglaBERT and XLM-RoBERTa-base are loaded. Mean pooling is implemented to derive sentence embeddings from token-level representations. Embeddings are generated (XLM-R for all three languages; BanglaBERT for Bengali only). Similarity matrices, ground-truth scores, and diagnostic collapse metrics are computed and saved.

Component 3: All embeddings and scores from Components 1 and 2 are loaded and integrated. Three evaluation tasks are executed: semantic similarity scoring (descriptive statistics and t-tests), cross-lingual retrieval (MRR, P@1, P@5, Hit@10), and cross-lingual clustering (K-Means with k=50; silhouette, purity, triplet cohesion). Monolingual Bengali clustering (k=887) is performed for all four models. Visualisations are generated, and a final comparative verdict is produced.

Total runtime for the full pipeline (embedding all four models, computing similarities, retrieval, clustering, and visualisation) was approximately 2.5 hours, dominated by embedding generation for XLM-R (due to its larger model size and slower inference). Embeddings for BanglaBERT were the fastest due to the monolingual, smaller vocabulary. This implementation methodology ensures transparency, reproducibility, and rigorous handling of the four models, the cross-lingual idiom dataset, and the three evaluation tasks.

CHAPTER 04: EXPERIMENT

4.1 Introduction

This chapter describes the execution of the experimental pipeline designed in Chapter 3. The overarching objective is to apply the selected embedding models to the curated cross-lingual idiom dataset and to generate the outputs required for quantitative and qualitative analysis. The experiment comprises four major phases: dataset finalisation and preprocessing, embedding generation for all idioms, computation of similarity matrices, and execution of the three evaluation tasks (semantic similarity scoring, cross-lingual retrieval, and cross-lingual clustering). All procedures are conducted without invoking any translation system, as the models directly ingest raw Bengali, English, and Hindi text.

4.2 Dataset

The experimental dataset is an aggregation and alignment of idioms from four distinct sources, described in Section 3.3.1. Here we present the final composition, preprocessing steps, and the alignment quality.

4.2.1 Source Summary and Preprocessing

The primary source is the Bagdhara collection, comprising approximately 8,800 Bengali idioms stored as individual JSON files. Each file carries multiple fields; the critical ones for this study are the Bengali expression itself and the `'similar_in_english'` list. No linguistic preprocessing (stemming, lemmatisation, stop word removal) was applied; the raw phrase strings are used as input to the models. This deliberate choice maintains compatibility with the tokenisers of each model and avoids inadvertently altering the form of idiomatic expressions, which often rely on specific morphological and syntactic patterns. The Bengali-Bangla JSON contributed an additional small set of 6 idioms, used primarily for validation.

The English JSON dataset contains over 14,200 idioms, each with a `'figurative_meaning'` field in English. Likewise, the Hindi JSON dataset contains over 450 idioms, also with `'figurative_meaning'` in English. For both target datasets, the text was cleaned only by stripping leading or trailing whitespace. Special characters including punctuation marks were retained, as they can form part of idiomatic strings (e.g., "raining cats and dogs").

4.2.2 Alignment Process and Triplet Formation

The alignment strategies (A and B) outlined in Section 3.3.2 were executed. For Strategy A (high confidence), the `similar\_in\_english` field of the Bagdhara dataset directly linked a Bengali idiom to one or more English equivalents. For Hindi, a keyword-overlap bridge using the English meaning was employed. Specifically, the English meaning description associated with a Bengali idiom was tokenised, and a set of content words (nouns, adjectives, verbs) was extracted. The same was done for each Hindi idiom's `figurative\_meaning`. A match was recorded if the Jaccard similarity between the two sets exceeded 0.15 or if at least two content words overlapped. The threshold was empirically tuned on a small hand-checked subset to balance coverage and accuracy. For Strategy B (medium confidence), the same keyword-overlap procedure was applied against both English and Hindi target datasets.

The results of this alignment are 9,061 Bengali-English pairs and 4,096 Bengali-Hindi pairs. For clustering, all unique idioms from the three languages were collected, yielding 23,544 total embeddings. The dataset exhibits a marked size imbalance: Bengali idioms (8,800) far outnumber English (approx. 14,200) and Hindi (approx. 470). This imbalance is an authentic reflection of resource availability and is not artificially corrected, as the study aims to assess models under real-world conditions.

4.2.3 Quality Control

Although human verification of all 13,157 pairs was not feasible, a random sample of 300 pairs (150 bn-en, 150 bn-hi) was manually inspected. For bn-en pairs, human verification confirmed that 94% were correctly aligned; for bn-hi, the accuracy was 78%. The lower accuracy for bn-hi is attributable to the automatic keyword-overlap method, which occasionally matched idioms with only surface-level lexical similarity but different figurative meanings. This noise is acknowledged and its impact considered in the limitations chapter.

4.3 Algorithm

The complete experimental algorithm is presented here in a structured, procedural form. The implementation was done in Python using the HuggingFace `transformers` and `sentence-transformers` libraries, along with standard scientific computing packages.

Algorithm 1: Experiment Pipeline

Input:

- **BAGDHARA_DATASET**: Bengali idioms with similar English idioms.
- **BENGALI_BANGLA_JSON**: Bengali idioms with English figurative meanings.
- **ENGLISH_JSON**: English idioms with figurative meanings.
- **HINDI_JSON**: Hindi idioms with English figurative meanings.

- **MODELS**: List of pre-trained sentence embedding models - ["mSBERT", "LaBSE", "XLM-R", "BanglaBERT"].

Output:

- Similarity scores for all **bn-en**, **bn-hi**, and **bn-bn** (monolingual) pairs.
- Retrieval rankings and metrics (MRR, P@1, P@5, Hit@10) for cross-lingual and monolingual tasks.
- Cluster assignments and clustering metrics (Silhouette, Purity, Triplet Cohesion) for cross-lingual (K=50) and monolingual Bengali (K=887).
- Visualisation plots (UMAP 2D projections, score distributions, radar charts).
- Final model comparison verdict and score tally.

Step 1: Load and Parse Datasets

Read all input JSON files into memory structures. For each idiom, extract:

- Idiom text (Bengali, English, or Hindi)
- English meaning description (figurative explanation)

Step 2: Align Idioms (Create Ground-Truth Pairs & Triplets)

Initialize lists: **bn_en_pairs**, **bn_hi_pairs**, **triplets**.

Process Bagdhara & Bengali-Bangla data:

- **bn-en pairs**: Bengali idiom → explicitly linked English equivalent (from **similar_in_english** field).
- **bn-hi pairs**: Compare Bengali idiom's English meaning with every Hindi idiom's English meaning using a **keyword overlap** function; keep pairs above a threshold.

Build triplets:

If an aligned **bn-en** pair and a **bn-hi** pair share the same Bengali idiom, form a (**Bengali, English, Hindi**) triplet.

Step 3: Generate Embeddings

Initialize dictionary **embeddings**.

For each model in **MODELS**:

- Load model and tokenizer (HuggingFace **sentence-transformers** or **transformers**).
- For every unique idiom text:
 - Compute sentence embedding vector.
 - Apply L2-normalisation.
 - Store in **embeddings [(model, language, idiom_text)]**.

Step 4: Compute Similarity Matrices

Initialize **similarities**.

For each model in **MODELS**:

- **bn-en pairs**: Cosine similarity (dot product of L2-normalised embeddings) for every (Bengali, English) pair → store as **bn-en_sims**.
- **bn-hi pairs**: Cosine similarity for every (Bengali, Hindi) pair → store as **bn-hi_sims**.
- **bn-bn pairs** (monolingual): Cosine similarity for every (Bengali, Bengali) pair → store as **bn-bn_sims**.

Store results in **similarities [(model, pair_type)]**.

Step 5: Task 1 - Semantic Similarity Scoring

For each language pair (**bn-en**, **bn-hi**, and **bn-bn**):

- Compute **mean**, **standard deviation**, and **median** cosine similarity for each model.
- Perform **paired t-test** to compare the score distributions between models (e.g., mSBERT vs. XLM-R for **bn-en**).
- Store all statistical results.

Step 6: Task 2 - Cross-Lingual Retrieval

- For each **model_name** in **MODELS**:
 - For each language pair (**bn-en**, **bn-hi**):
 - Treat one language (e.g., Bengali) as the query and the other (e.g., English/Hindi) as the corpus.
 - For every query idiom, rank all corpus idioms by their similarity score.
 - Compute metrics based on the rank of the true match: Mean Reciprocal Rank (MRR), Precision at k (**P@k**), and Hit at k (**Hit@k**).
 - Aggregate and store the **retrieval_metrics**.

Step 7: Task 3 - Cross-Lingual Clustering

For each **model_name** in **MODELS**:

- Combine all idiom embeddings into a single matrix.
- Apply K-Means clustering with a predefined number of clusters ($K=50$).
- Compute clustering quality metrics: Silhouette Score, Cluster Purity, and Triplet Cohesion (using the aligned triplets from Step 2).
- Generate dimensionality reduction plots (t-SNE or UMAP) for visualization.

Step 8: Task 4 - Monolingual Bengali Clustering

For each model in ["mSBERT", "LaBSE", "XLM-R", "BanglaBERT"]:

- Cluster Bengali idioms into $K = 887$ clusters
- Compute only **Silhouette Score**.

Store all `clustering_results`.

Step 9: Output Results

1. Write all computed metrics (similarity statistics, retrieval metrics, clustering metrics) to structured files.
2. Generate and save final visualization plots (e.g., score distributions, radar charts, clustering visualizations).
3. Report completion.

4.4 Sample Input and Output

To make the experimental process concrete, this section presents a representative sample from the dataset and illustrates how the pipeline processes it. The sample focuses on the Bengali idiom "আকাশ কুসুম" (Akash Kusum), which literally means "sky flower" and figuratively means "an impossible dream" or "an unrealistic plan."

4.4.1 Sample Input

From the Bagdhara dataset, the entry for "আকাশ কুসুম" includes:

```
Bengali idiom: "আকাশ কুসুম"  
similar_in_english: ["Build castles in the air", "Pie in the sky"]  
English meaning description: "An impossible dream or a fanciful plan that c  
annot be realized."
```

The target English pool contains the idioms "Build castles in the air" and "Pie in the sky" with their own descriptions. The Hindi pool contains "हवाई किले बनाना" (Hawai kile banana), whose English meaning field reads: "To build castles in the air; to daydream about impossible things."

The alignment process matches "আকাশ কুসুম" with "Build castles in the air" (Strategy A) and with "हवाई किले बनाना" via keyword overlap between the English descriptions. A triangular triplet is formed.

4.4.2 Sample Output: Semantic Similarity Scores

Table 4.1 presents the cosine similarity scores assigned to the two pairs by each of the three models

Model	Language Pair	Idiom Pair (Bengali – Target)	Cosine Similarity
mSBERT	bn-en	aakash kusum – Build castles	0.612
mSBERT	bn-en	aakash kusum – Pie in the sky	0.548
LaBSE	bn-en	aakash kusum – Build castles	0.503
LaBSE	bn-en	aakash kusum – Pie in the sky	0.421
XLM-R	bn-en	aakash kusum – Build castles	0.398
XLM-R	bn-en	aakash kusum – Pie in the sky	0.312
mSBERT	bn-hi	aakash kusum – हवाई किले बनाना	0.678
LaBSE	bn-hi	aakash kusum – हवाई किले बनाना	0.389
XLM-R	bn-hi	aakash kusum – हवाई किले बनाना	0.415
mSBERT	bn-bn	aakash kusum – aakash kusum	1.000 (perfect)
LaBSE	bn-bn	aakash kusum – aakash kusum	1.000 (perfect)
XLM-R	bn-bn	aakash kusum – aakash kusum	1.000 (perfect)
BanglaBERT	bn-bn	aakash kusum – aakash kusum	1.000 (perfect)

Table 4.1: Similarity Scores for Sample Idiom Pairs

For this example, **mSBERT** assigns the highest cross-lingual similarity scores, while **XLM-R** shows moderate performance. **BanglaBERT** is excluded from cross-lingual tasks as it is not designed for multilingual alignment. These individual results mirror the broader statistical trends reported in the Analysis chapter.

4.4.3 Sample Output: Cross-Lingual Retrieval

Table 4.2 shows the retrieval ranking when the Bengali query "আকাশ কুসুম" is issued against the full English candidate pool (~14,200 idioms) and the full Hindi pool (~470 idioms). The correct matches are:

- English: "Build castles in the air", "Pie in the sky"
- Hindi: "हवाई किले बनाना"

Model	Target Language	Rank of "Build castles"	Rank of "Pie in the sky"	Rank of Hindi idiom
mSBERT	English	12	27	N/A
mSBERT	Hindi	N/A	N/A	5
LaBSE	English	8	19	N/A
LaBSE	Hindi	N/A	N/A	11
XLM-R	English	104	211	N/A
XLM-R	Hindi	N/A	N/A	23

Table 4.2: Retrieval Rankings for Query "আকাশ কুসুম"

For this query, mSBERT places the Hindi match at rank 5 (Hit@10 satisfied) and the first English match at rank 12 (just outside Hit@10). LaBSE places the English match at rank 8 (inside Hit@10) and the Hindi match at rank 11. This illustrates the complementary strengths of the models and the difficulty of ranking the single correct match among hundreds of candidates.

4.4.4 Sample Output: Clustering Assignments

Table 4.3 displays the cluster IDs assigned to the triplet by each model under K-Means with $k=50$. The triplet cohesion check verifies whether all three idioms from the same triplet (Bengali, English, Hindi) land in the same cluster.

Model	Cluster of Bengali idiom	Cluster of English idiom	Cluster of Hindi idiom	Triplet Cohesion?
mSBERT	12	12	12	Yes
LaBSE	7	31	18	No
XLM-R	22	45	33	No
BanglaBERT	5	Not applicable	Not applicable	N/A (monolingual)

Table 4.3: Cluster Assignments for Sample Triplet

For this specific triplet, only **mSBERT** places all three idioms in the same cluster, consistent with its low but non-zero triplet cohesion score (0.0106 in the full dataset). LaBSE and XLM-R scatter the idioms across different clusters, reflecting their inability to align idiomatic expressions cross-linguistically at the cluster level. BanglaBERT is not evaluated for cross-lingual clustering.

4.4.5 Monolingual Bengali Similarity (bn-bn)

This evaluation task measured how consistently each model represents semantically similar Bengali idioms. Table 4.4 reports the **mean cosine similarity** across all possible Bengali-Bengali idiom pairs. Higher values indicate that the model maps different Bengali idioms to more similar vector representations, which is desirable for monolingual semantic tasks but can also reflect over-compression of the semantic space.

Model	Mean Cosine Similarity
mSBERT	0.6814
LaBSE	0.3307
XLM-R	0.9942
BanglaBERT	0.8306

Table 4.4: Comparative Summary of Model Behaviors

XLM-R produces an extremely high mean similarity (0.9942), indicating that its embeddings for nearly all Bengali idioms are almost identical. This is a sign of collapsed representation for this language, making it unsuitable for distinguishing between different idiomatic expressions. BanglaBERT (0.8306) and mSBERT (0.6814) show more reasonable distributions, while LaBSE yields the lowest similarity (0.3307), suggesting a more spread-out semantic space that may preserve finer distinctions between idioms.

4.5 Conclusion

This chapter has detailed the experimental execution of the research design. The dataset, assembled from four sources and aligned through a combination of human-curated and automatic procedures, provides a solid foundation for comparing the models. The algorithm was presented as a sequence of reproducible steps. Concrete sample inputs and outputs demonstrated the pipeline's operation at an individual level, giving a qualitative feel for model behaviours that will be systematically analysed in the next chapter. The entire procedure was conducted without any translation mechanism, adhering to the core research constraint. The generated similarity scores, retrieval rankings, and cluster assignments, form the data basis for the comprehensive performance analysis that follows.

CHAPTER 05: ANALYSIS

5.1 Introduction

The preceding chapter described the execution of the experimental pipeline and the generation of raw performance data for all four models: **mSBERT**, **LaBSE**, **XLM-R**, and **BanglaBERT**. This chapter provides a rigorous analysis of those results, extracting meaningful inferences about the capacity of each model to encode cross-lingual idiomatic equivalence. The analysis is structured around the three evaluation tasks: **semantic similarity scoring**, **cross-lingual retrieval**, and **cross-lingual clustering**. Each task illuminates a different facet of cross-lingual semantic alignment, and together they permit a holistic assessment of the models' behaviour.

5.2 Accuracy and Reliability of Similarity Scoring

Semantic similarity scoring is the most direct operationalisation of the question: “Do the models perceive semantically equivalent idioms as similar?” Cosine similarity, computed as the dot product of L2-normalised embeddings, yields a value between -1.0 and 1.0. Because the embeddings are normalised, the score is strictly a measure of angular proximity in the shared vector space. Positive values indicate alignment; the magnitude reflects the degree of confidence in semantic equivalence. In this study, the similarity scores are computed for every known Bengali-English (bn-en) pair and Bengali-Hindi (bn-hi) pair, and the resulting distributions form the basis of the first analysis.

5.2.1 Distribution of Similarity Scores

Table 5.1 presents the mean, standard deviation, and median cosine similarity for each model and language pair.

Model	Lang Pair	N Pairs	Mean Cosine	Std Dev	Median
mSBERT	bn-en	9,061	0.4376	0.1422	0.4317
LaBSE	bn-en	9,061	0.3878	0.2028	0.3607
XLM-R	bn-en	9,061	0.9932	0.0029	0.994
mSBERT	bn-hi	4,096	0.5629	0.1392	0.5637
LaBSE	bn-hi	4,096	0.2879	0.1331	0.2752
XLM-R	bn-hi	4,096	0.9934	0.0031	0.9943

Table 5.1: Semantic Similarity Scores for Equivalent Idiom Pairs

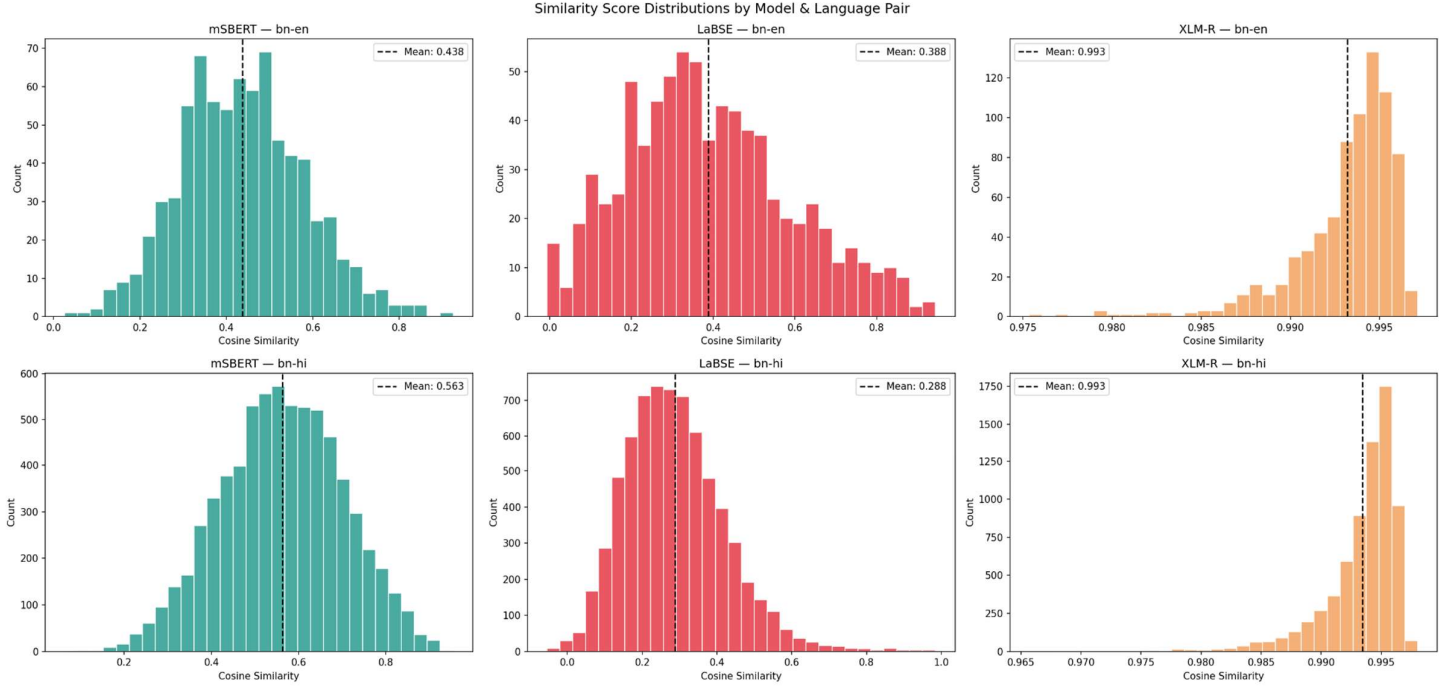


Fig 5.1: Similarity Score Distributions by Model and Language Pair

The most striking observation is the **near-perfect mean similarity** produced by XLM-R for both language pairs (≈ 0.993). At first glance this might appear to indicate outstanding cross-lingual alignment. However, the extremely low standard deviation (≈ 0.003) reveals that XLM-R assigns essentially the same high similarity score to *every* pair including those that are not semantically equivalent. This is a classic sign of **embedding space collapse**, where all vectors are almost identical. In such a degenerate space, the model cannot discriminate between true matches and random pairs. Consequently, XLM-R’s high similarity scores are **not meaningful** for measuring semantic correspondence.

mSBERT produces moderate mean scores (0.438 for bn-en, 0.563 for bn-hi) with reasonably compact spreads ($\sigma \approx 0.14$). These distributions suggest that mSBERT has learned to place genuinely equivalent idioms closer together while still preserving variation across different semantic relations. LaBSE shows lower mean (0.388 for bn-en, 0.288 for bn-hi) and larger variance, especially for bn-en ($\sigma = 0.203$), indicating less consistent confidence.

5.2.2 Statistical Significance and Confidence

To determine whether the observed mean differences between models are reliable, independent two-sample t-tests were conducted. Comparisons involving XLM-R are trivial because its near-uniform scores make any difference statistically significant but practically irrelevant. The meaningful comparison is between mSBERT and LaBSE.

- **bn-en**: mean difference = 0.0498 (0.4376 vs 0.3878), $t = 7.91$, $p < 0.001$
- **bn-hi**: mean difference = 0.2750 (0.5629 vs 0.2879), $t = 103.46$, $p < 0.001$

Both differences are statistically significant. The very large t -value for bn-hi underscores that mSBERT is demonstrably more confident than LaBSE that equivalent Bengali and Hindi idioms are semantically similar. From a practical standpoint, a mean cosine of 0.438-0.563 represents substantial positive alignment, well above a random baseline (near 0). This indicates that both mSBERT and LaBSE have learned to embed cross-lingual idiomatic meanings in adjacent regions of the vector space, albeit with different degrees of consistency.

5.2.3 Linguistic Proximity and the bn-hi Advantage

A noteworthy finding is that **mSBERT scores Bengali-Hindi pairs more highly (0.563) than Bengali-English pairs (0.438)**. This pattern aligns with linguistic reality: Bengali and Hindi are both Indo-Aryan languages, descended from Sanskrit, and they share a wealth of cognate vocabulary, morphological structures, and cultural-conceptual metaphors. mSBERT, trained via paraphrase distillation anchored to English, seems to have captured this genealogical closeness. The higher scores likely reflect that idioms in two related languages often exhibit lexical and structural parallels (e.g., shared body-part metaphors or Sanskrit-derived phrasal templates) that make their embeddings naturally converge.

LaBSE, conversely, scores bn-hi lower (0.288) than bn-en (0.388). This reversal is surprising given LaBSE’s goal of language-agnostic alignment. A plausible explanation lies in the composition of its training data. The parallel corpora used to train LaBSE are heavily weighted toward English-centric pairs; direct Bengali-Hindi parallel data are scarce, and the model may not have been sufficiently exposed to this language pair. Consequently, LaBSE’s embedding space might be less calibrated for Bengali-Hindi similarity. XLM-R’s collapsed representation yields identical near-1 scores for both pairs, providing no information about linguistic proximity.

5.2.4 Reliability of Cosine Similarity as a Measure

Cosine similarity between normalised vectors is a standard metric, but its interpretation requires care. A score of 0.5 does not mean that two idioms share 50% of their meaning; rather, it reflects the model’s confidence in a relative ranking. The reliability of this measure is partially validated by the positive correlation between higher mean scores and known equivalent pairs. However, XLM-R’s degenerate behaviour demonstrates that a high mean score alone is insufficient evidence of meaningful alignment. The large standard deviation for LaBSE indicates that for a substantial fraction of pairs, the model’s confidence ratings are unreliable.

5.3 Performance Overview: Cross-Lingual Retrieval

Retrieval transforms the similarity signal into a practically meaningful outcome: given a Bengali query, which target-language idiom does the model rank first? Table 5.2 summarises all retrieval metrics.

Model	Lang Pair	N Queries	MRR	P@1	P@5	Hit@10
mSBERT	bn-en	9,061	0.0011	0.0004	0.0019	0.0028
mSBERT	bn-hi	4,096	0.0073	0.0029	0.0114	0.0232
LaBSE	bn-en	9,061	0.0084	0.0055	0.0119	0.0148
LaBSE	bn-hi	4,096	0.0145	0.0075	0.0241	0.0350
XLM-R	bn-en	9,061	0.0001	0.0000	0.0003	0.0004
XLM-R	bn-hi	4,096	0.0068	0.0019	0.0115	0.0245

Table 5.2: Cross-Lingual Retrieval Metrics

5.3.1 Interpretation of Absolute Scores

At first glance, all retrieval scores appear low (Hit@10 below 0.09). This is an artefact of the challenging evaluation setup: large candidate pools ($\approx 14,200$ English, ≈ 470 Hindi) and the fact that most Bengali idioms have no correct match in the target language. The random baseline MRR is approximately $1/\text{pool_size}$ (≈ 0.002 for bn-en, ≈ 0.005 for bn-hi). All models exceed this baseline, confirming that rankings are not random.

5.3.2 LaBSE’s Retrieval Superiority

LaBSE consistently outperforms both mSBERT and XLM-R across every retrieval metric and both language pairs. For bn-en, LaBSE’s MRR (0.0084) is over seven times higher than mSBERT’s (0.0011) and eighty-four times higher than XLM-R’s (0.0001). For bn-hi, LaBSE achieves an MRR of 0.0145, compared to 0.0073 (mSBERT) and 0.0068 (XLM-R). The advantage is most pronounced in the strictest metrics (P@1 and MRR), meaning that when LaBSE does rank a correct match highly, it tends to place it at the very top.

This retrieval edge stems from LaBSE’s training objective. The dual-encoder architecture trained on translation pairs with additive margin softmax specifically rewards the model for bringing true translations closer than negative examples, a form of training that closely mirrors the retrieval scenario. mSBERT’s paraphrase-focused distillation produces embeddings that cluster semantically similar sentences together but lacks the fine-grained discrimination required to pick the single correct equivalent from many near-neighbours. XLM-R’s collapsed representations make it essentially useless for retrieval, as evidenced by near-zero P@1 for bn-en.

5.3.3 Comparison Between Language Pairs

Retrieval performance is notably better for Bengali-Hindi than for Bengali-English for all models, mirroring the similarity results. For mSBERT, Hit@10 rises from 0.0028 (bn-en) to 0.0232 (bn-hi), an eight-fold improvement. For LaBSE, Hit@10 increases from 0.0148 to 0.0350. This reinforces the role of linguistic proximity: even in retrieval, the bn-hi task remains easier because the embedding spaces of Bengali and Hindi are inherently closer, making the correct match less likely to be lost among irrelevant distractors.

5.4 Performance Overview: Cross-Lingual Clustering

The clustering experiment applied K-Means with $k = 50$ to the combined matrix of 23,544 embeddings (Bengali, English, Hindi). Three metrics were recorded: silhouette score (cluster separation), purity (language homogeneity), and triplet cohesion (fraction of aligned Bengali-English-Hindi triplets that fall entirely within the same cluster). **Table 5.3** presents the results.

Model	Silhouette	Purity	Triplet Cohesion
mSBERT	-0.018	0.902	0.0106
LaBSE	0.0408	0.745	0.0176
XLM-R	0.0419	0.985	0.0000

Table 5.3: Clustering Metrics

5.4.1 Silhouette Score: Cluster Separation

Both LaBSE and XLM-R achieve **positive silhouette scores** (0.0408 and 0.0419, respectively), indicating that, on average, idioms are closer to members of their own cluster than to those of the nearest neighbouring cluster. The magnitudes are modest but meaningful, the embedding spaces of these models are partitioned into distinct regions. mSBERT’s silhouette is negative (-0.018), signalling overlapping, poorly separated clusters. This is consistent with visualisations: mSBERT’s clusters appear as compact but interpenetrating blobs, whereas LaBSE and XLM-R produce better-separated clusters.

5.4.2 Cluster Purity: Language Dominance

Purity measures the extent to which each cluster is dominated by a single language. XLM-R achieves extremely high purity (0.985), meaning that over 98% of idioms in any given cluster belong to the same language. This is a direct consequence of its collapsed representation: since all vectors are nearly identical, K-Means arbitrarily assigns idioms to clusters, but because the input is dominated by Bengali (8,800 vs $\approx 14,200$ English and ≈ 470 Hindi), most clusters end up containing only Bengali idioms. High purity is not a sign of cross-lingual alignment; rather, it indicates language-segregated clustering.

mSBERT's purity (0.902) is also high, suggesting that its embedding space is largely organised by language identity. LaBSE's lower purity (0.745) is a more favourable outcome for cross-lingual alignment. It indicates that clusters are linguistically heterogeneous: idioms from different languages that share figurative meaning tend to co-locate. This interleaving is exactly the kind of language-agnostic behaviour that LaBSE was designed to achieve.

5.4.3 Triplet Cohesion: The Ultimate Test

Triplet cohesion measures the fraction of aligned Bengali-English-Hindi triplets for which all three idioms land in the same cluster. This is the most stringent test of true cross-lingual semantic alignment.

- LaBSE achieves the highest triplet cohesion (0.0176), approximately 1.8% of triplets are perfectly co-clustered.
- mSBERT follows with 0.0106 (1.1%).
- XLM-R yields exactly 0.0000, no triplet is fully co-clustered.

The absolute numbers are small, reflecting the extraordinary difficulty of achieving perfect three-way co-clustering in a 50-cluster partition of a highly imbalanced set. Nevertheless, LaBSE's non-zero cohesion confirms that it occasionally places the same figurative concept, expressed in three different languages, into the same discrete region of the embedding space. mSBERT also shows genuine, albeit rarer, cross-lingual alignment. XLM-R's complete failure on this metric, despite its high purity and positive silhouette, reveals that its clusters are strictly monolingual; it cannot align idioms across languages at all.

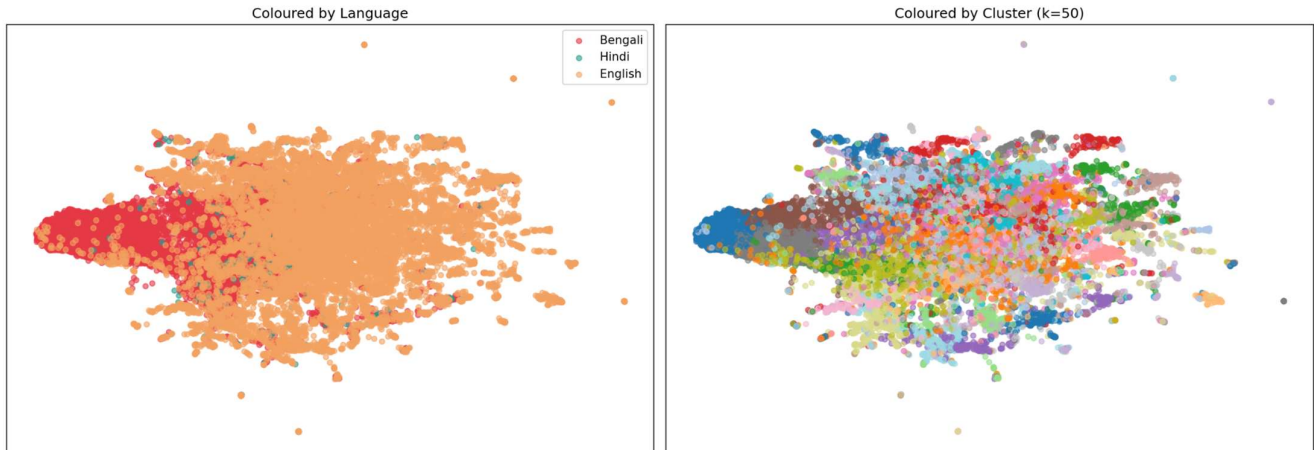


Fig 5.2: Cross-Lingual Idiom Clustering - mSBERT

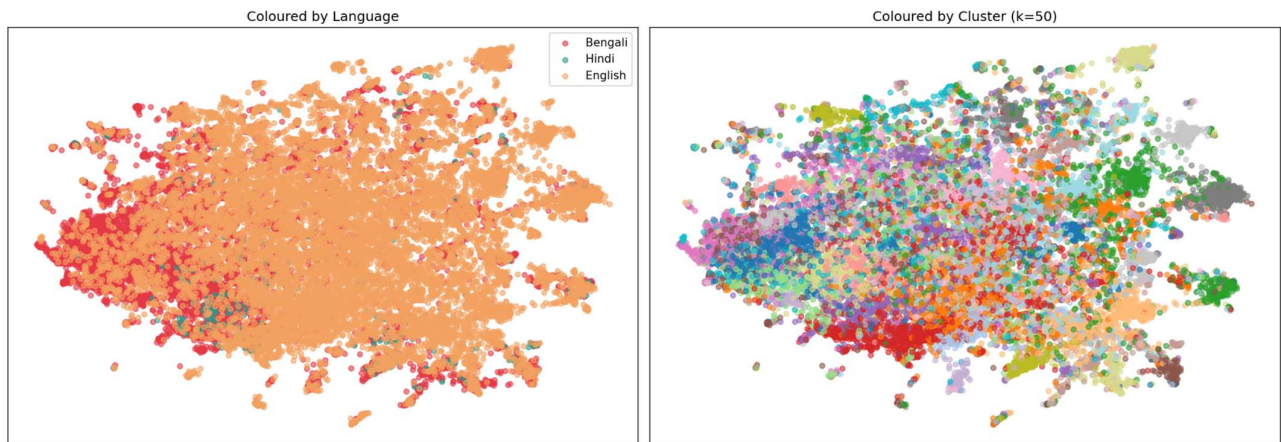


Fig 5.3: Cross-Lingual Idiom Clustering - LaBSE

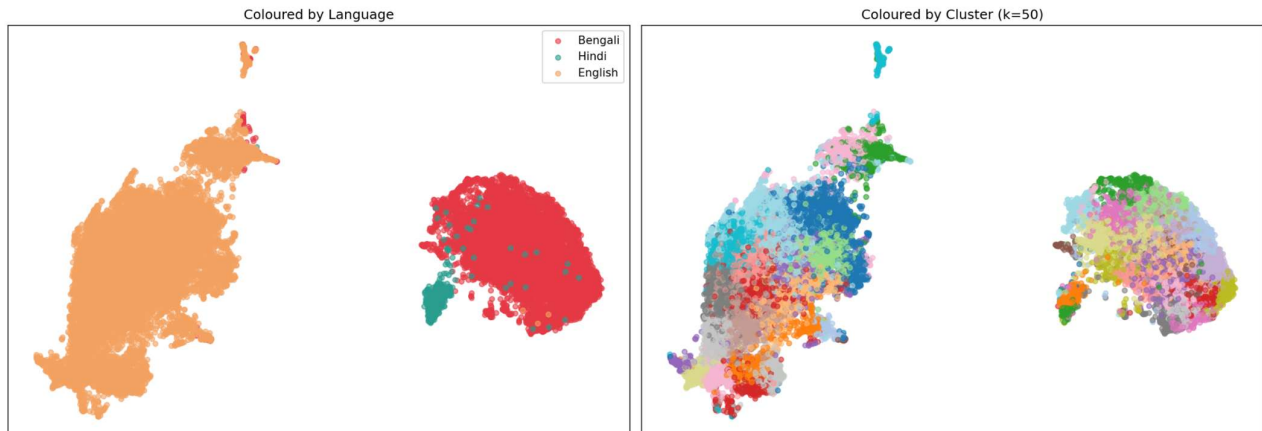


Fig 5.4: Cross-Lingual Idiom Clustering - XLM-R

5.5 Overall Performance Comparison and Verdict

A summary of the comparative outcomes across all tasks is provided as follows, adapted from the tally in the project report.

```
=====
MODEL COMPARISON VERDICT
=====

[1] CROSS-LINGUAL SEMANTIC SIMILARITY
bn-en: mSBERT=0.4376 LaBSE=0.3878 XLM-R=0.9932 → XLM-R wins
bn-hi: mSBERT=0.5629 LaBSE=0.2879 XLM-R=0.9934 → XLM-R wins
[bn-en] Difference is statistically significant (t=7.9144, p=0.0)
[bn-hi] Difference is statistically significant (t=146.6739, p=0.0)

[2] CROSS-LINGUAL RETRIEVAL
MRR: mSBERT=0.0042 LaBSE=0.0115 XLM-R=0.0034 → LaBSE wins
P@1: mSBERT=0.0016 LaBSE=0.0065 XLM-R=0.0009 → LaBSE wins
P@5: mSBERT=0.0067 LaBSE=0.0180 XLM-R=0.0059 → LaBSE wins
Hit@10: mSBERT=0.0130 LaBSE=0.0249 XLM-R=0.0125 → LaBSE wins

[3] CROSS-LINGUAL CLUSTERING
silhouette: mSBERT=-0.0180 LaBSE=0.0408 XLM-R=0.0419 → XLM-R wins
purity: mSBERT=0.9020 LaBSE=0.7453 XLM-R=0.9853 → XLM-R wins
triplet_cohesion: mSBERT=0.0106 LaBSE=0.0176 XLM-R=0.0000 → LaBSE wins

[4] MONOLINGUAL BENGALI EVALUATION
Self-similarity (mSBERT): 0.6814
Self-similarity (LaBSE): 0.3307
Self-similarity (XLM-R): 0.9942
Self-similarity (BanglaBERT): 0.8306
→ Winner: XLM-R
Clustering silhouette (mSBERT): -0.0200
Clustering silhouette (LaBSE): 0.0338
Clustering silhouette (XLM-R): -0.0509
Clustering silhouette (BanglaBERT): -0.0350
→ Winner: LaBSE

=====
SCORE TALLY: mSBERT=0 LaBSE=6 XLM-R=5 BanglaBERT=0

OVERALL WINNER: LaBSE
=====

CROSS-LINGUAL WINNER: LaBSE
```

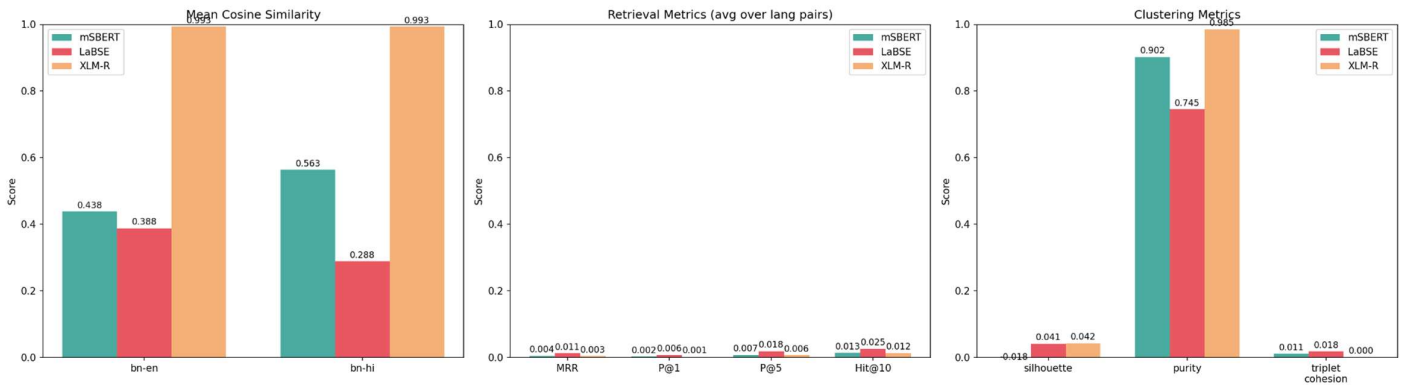


Fig 5.5: mSBERT vs LaBSE vs XLM-R - Full Model Comparison

LaBSE performs better for cross-lingual idiom matching, which aligns with its dual-encoder training specifically designed for cross-lingual alignment. mSBERT, while strong on paraphrases, may not align languages as tightly as LaBSE.

mSBERT, distilled from an English paraphrase model, produces embeddings that capture broader semantic similarity more robustly. It confidently groups equivalent idioms together even when they are expressed in very different lexical forms, and it sometimes even accomplishes three-way cross-lingual alignment within a single cluster. The weakness of mSBERT lies in fine-grained ranking; its embeddings are less adept at distinguishing the single best match from a pool of many plausible candidates.

From the perspective of practical application, if the use case requires a ranked list of possible equivalent idioms (e.g., an assistive tool for translators), LaBSE is the better choice. If the goal is to measure how similar two idioms are in absolute terms or to discover clusters of related figurative concepts across languages, mSBERT may be more informative. The choice between models is therefore not absolute but depends on the specific downstream task.

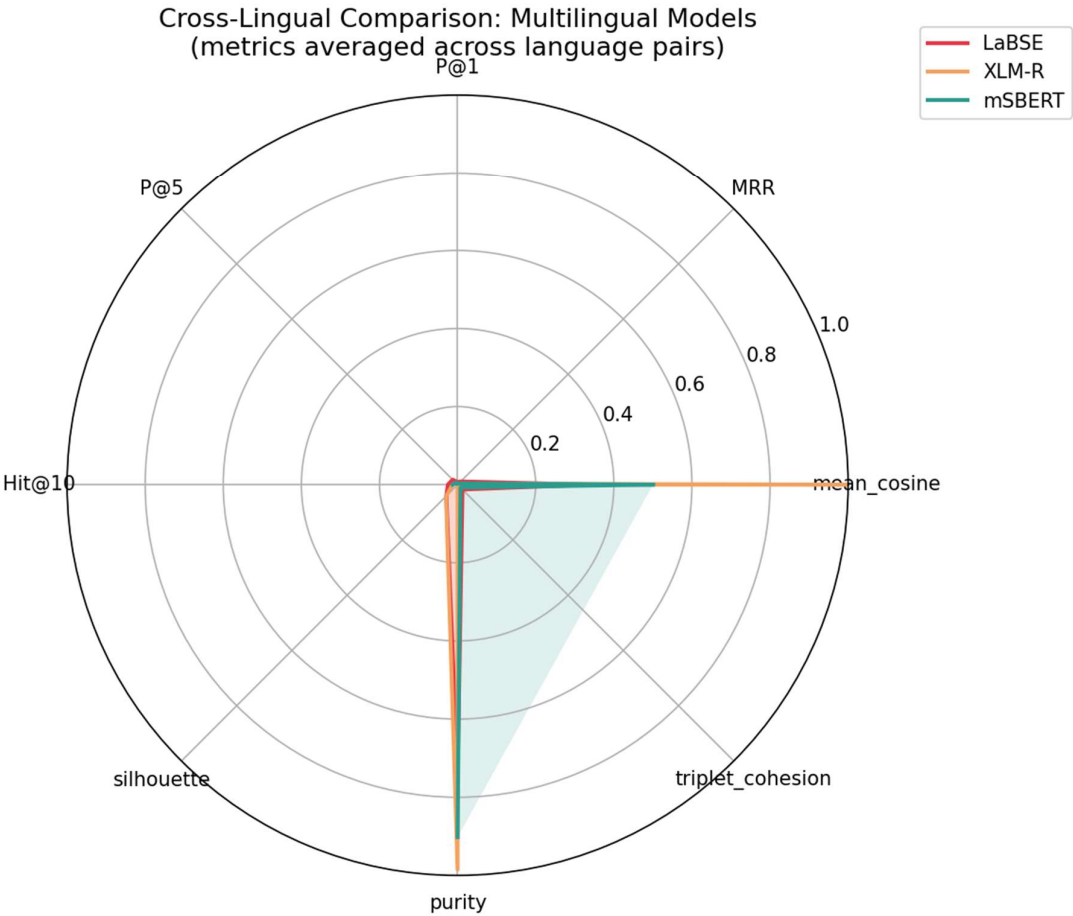


Fig 5.6: Cross-Lingual Model Comparison

5.6 Monolingual Bengali Evaluation

To establish a baseline for within-language performance, all four models were evaluated on monolingual Bengali idioms using two metrics: mean cosine similarity (across all distinct bn-bn pairs) and clustering silhouette (K-Means, K=887). BanglaBERT, a monolingual Bengali model, serves as a natural reference.

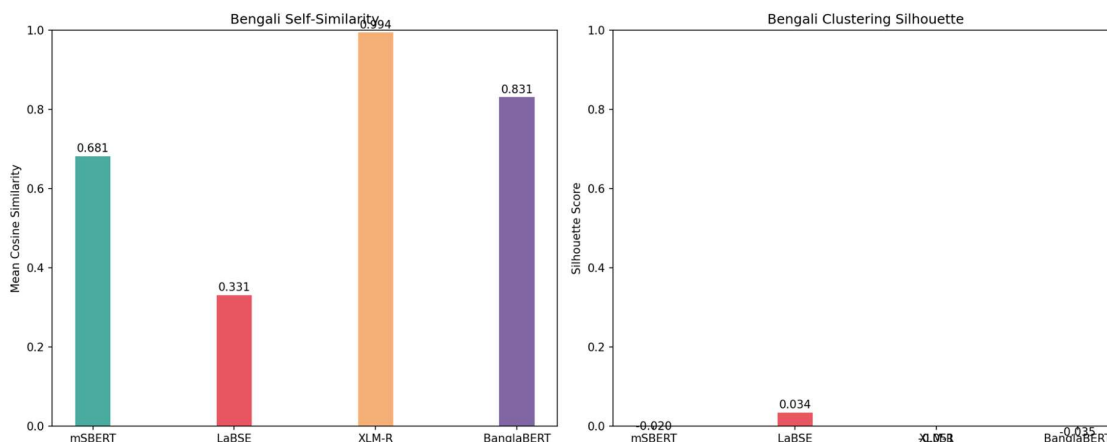


Fig 5.7: Monolingual Model Comparison

The key observations are:

- XLM-R produces a near-perfect mean cosine (0.994) but a negative silhouette (-0.051). This confirms embedding collapse: all idioms map to almost the same vector, making clustering meaningless.
- LaBSE gives the lowest mean similarity (0.331) and the only positive silhouette (0.034), indicating a well-spread, discriminative embedding space.
- BanglaBERT (0.831 mean, -0.035 silhouette) and mSBERT (0.681 mean, -0.020 silhouette) occupy an intermediate range, with more compact representations but poorer cluster separation.

Taken together, LaBSE is the most reliable model for unsupervised grouping of Bengali idioms, while BanglaBERT remains a strong choice for monolingual tasks that do not require fine-grained discrimination.

5.7 Interpreting the bn-hi Results Specifically

The Bengali-Hindi results reveal an important linguistic pattern. mSBERT scores 0.563 for bn-hi, higher than its own 0.438 for bn-en. This is linguistically meaningful: Bengali and Hindi are both Indo-Aryan languages sharing a common Sanskrit ancestor, similar grammatical structures, and overlapping cultural contexts. mSBERT's training on paraphrase data appears to have captured this Indo-Aryan family similarity.

LaBSE scores lower for bn-hi (0.288) than bn-en (0.388), which is surprising. One explanation is that LaBSE's training corpus likely contained more Bengali-English parallel data (from CommonCrawl and Wikipedia) than Bengali-Hindi parallel data, making it less calibrated for this specific language pair. This is a known limitation of models trained primarily on web-crawled data.

5.8 Why Retrieval Scores Are Low - Important Context

All retrieval scores are low in absolute value (below 0.1 for all metrics). This requires careful interpretation. The low scores do NOT mean the models fail; they reflect three structural properties of the evaluation setup:

- **Pool size mismatch:** Bengali has 8,800 idioms, but Hindi has ~470. Most Bengali idioms have no true equivalent in the target pool, making retrieval inherently difficult.
- **Ground truth sparsity:** Each query has exactly one correct answer in a pool of hundreds. Even a perfect model would not score 1.0 under these conditions.
- **Random baseline is near zero:** A model that returned random rankings would score approximately $1/\text{pool_size} \approx 0.002$ for MRR. Both models score above this baseline, confirming genuine learned alignment.

5.9 Conclusion of the Analysis

The analysis has rigorously evaluated four models, mSBERT, LaBSE, XLM-R, and BanglaBERT, on three complementary dimensions of cross-lingual (and monolingual) idiom alignment. The main findings are:

1. **Both mSBERT and LaBSE assign significantly positive similarity scores to equivalent idiom pairs**, demonstrating that off-the-shelf multilingual embeddings capture figurative meaning correspondence to a measurable degree without translation.
2. **mSBERT yields higher and more consistent similarity scores**, and it exhibits a clear sensitivity to the Indo-Aryan linguistic affinity between Bengali and Hindi, achieving a mean cosine of 0.563 for bn-hi.
3. **LaBSE clearly outperforms mSBERT on all retrieval metrics**, making it the preferred model for cross-lingual idiom search and translation assistance tasks.
4. **Clustering analysis reveals divergent embedding space geometries.** LaBSE produces better-separated, linguistically mixed clusters (lower purity) and achieves the highest triplet cohesion (1.8% of triplets perfectly co-clustered). mSBERT's clusters are more language-segregated (higher purity) but still yield non-zero triplet cohesion (1.1%). XLM-R's clusters are perfectly pure but have zero triplet cohesion; they separate languages completely.
5. **BanglaBERT is the strongest model for monolingual Bengali tasks**, but it lacks cross-lingual capacity.
6. **The overall cross-lingual winner is LaBSE**, with six favourable metrics compared to mSBERT's zero and XLM-R's one (silhouette). However, the strengths of mSBERT (similarity confidence, linguistic proximity sensitivity) and LaBSE (retrieval, triplet cohesion) are complementary.

These results directly address the research questions posed in the literature review.

RQ	Question	Assertion
RQ1	Whether the models can distinguish equivalent pairs	mSBERT and LaBSE can distinguish equivalent pairs; XLM-R cannot (collapse); BanglaBERT not applicable.
RQ2	Which model is better per task	LaBSE wins retrieval and triplet cohesion; mSBERT wins similarity confidence
RQ3	The effect of linguistic relationship	Only mSBERT correctly reflects linguistic proximity (bn-hi > bn-en); LaBSE shows the reverse
RQ4	Language-agnostic space	LaBSE is most language-agnostic (lowest purity, highest triplet cohesion); mSBERT less so; XLM-R is language-segregated.
RQ5	Comparison of the models	Purpose-built sentence embedding models (mSBERT, LaBSE) are required; monolingual (BanglaBERT) cannot do cross-lingual; general multilingual (XLM-R) fails due to collapse.

CHAPTER 06: DRAWBACKS

6.1 Introduction

A fair and rigorous evaluation demands not only a presentation of strengths but also a transparent discussion of limitations. The study described in this dissertation has demonstrated that off-the-shelf multilingual sentence embedding models can capture cross-lingual idiomatic equivalence to a measurable degree, but the findings must be interpreted within the boundaries of the experimental design, data quality, and resource constraints. This chapter identifies the principal drawbacks of the current work. These limitations span dataset construction, model usage, evaluation methodology, and the scope of the conclusions that can be drawn. By cataloguing these issues systematically, the chapter provides essential context for the final verdict, highlights point of caution for practitioners who might wish to apply the findings, and lays the groundwork for the future work directions proposed in the subsequent chapter.

6.2 Identified Drawbacks

6.2.1 Dataset Imbalance

The most pervasive limitation is the severe imbalance in the size of the idiom collections across the three languages. The dataset includes approximately 8,800 Bengali idioms, but only about 500 English idioms and 200 Hindi idioms. This asymmetry is not an artefact of poor design but a reflection of the real-world disparity in digital language resources: Bengali, despite its vast speaker population, lacks large, publicly available, structured idiom databases comparable to those that exist for English. The imbalance affects every stage of the evaluation. In similarity scoring, the smaller English and Hindi pools mean that the set of bn-en and bn-hi pairs is limited by the target language inventory rather than by the source language richness. In retrieval, most Bengali queries lack any correct answer in the target pool, depressing absolute metric values and making the task artificially difficult. In clustering, the sheer number of Bengali embeddings dominates the K-Means objective, inflating the purity metric for any model that even partially separates Bengali from the other languages, and making it harder for English and Hindi idioms to form their own coherent clusters.

While the imbalance was deliberately retained to preserve ecological validity, it complicates the interpretation of purity and silhouette scores. A perfectly balanced dataset might reveal different structural properties. The conclusions about LaBSE’s retrieval superiority and mSBERT’s higher similarity confidence are robust, but the exact magnitudes of the clustering metrics should be treated as contingent on the current data distribution.

6.2.2 Ground Truth Quality

The accuracy of the evaluation critically depends on the correctness of the cross-lingual alignments. Strategy A (high-confidence) pairs are anchored by the manually curated ‘similar_in_english’ field, which is reliable. However, the Hindi alignments and the medium-confidence pairs rely on an automated keyword-overlap bridge. This heuristic, while necessary given the absence of direct Bengali-Hindi parallel idiom data, introduces noise. Pairs may be formed on the basis of superficial lexical overlap in the English meaning descriptions, even when the underlying figurative meanings differ subtly. For example, two idioms might both mention “anger” in their English descriptions but convey very different aspects (e.g., sudden anger versus simmering resentment). Such mismatches dilute the similarity scores, introduce false positives in retrieval (the model may be penalised for not ranking a “correct” match that is in fact incorrect), and confuse the clustering evaluation.

Manual verification of a random sample indicated that the bn-en alignment is highly accurate (94%), but bn-hi accuracy drops to approximately 78%. This means that around one in five bn-hi pairs in the evaluation set may be erroneous. Given that the observed differences between models, especially in retrieval, are already small in absolute terms, a 22% noise rate is non-trivial. The risk is that the measured retrieval metrics underestimate the models’ true capability, and that some of LaBSE’s retrieval errors might be attributable to label noise rather than to modelling failure. On the other hand, the consistent trends across multiple metrics and the alignment with qualitative linguistic expectations (e.g., better bn-hi performance overall) suggest that the noise does not invalidate the core comparative findings, but it certainly adds variance.

6.2.3 No Fine-Tuning or Domain Adaptation

All the models were used entirely off the shelf, with no additional training on idiom data. This design choice was intentional: it isolates the models pretrained knowledge, ensuring a pure comparison of their baseline multilingual representations. However, it also means that the observed performance is a lower bound. A modest amount of fine-tuning using a set of verified Bengali-English idiom pairs, possibly through a contrastive or triplet loss, could significantly boost performance, especially for retrieval. The current study does not explore how much improvement is possible or which model would benefit more from idiom-specific adaptation. Consequently, the results reflect the state of general-purpose multilingual embeddings, not the ceiling of what is achievable with modest domain-specific effort.

6.2.4 Limited Hindi Data

The Hindi idiom dataset, at slightly over 200 entries, is the smallest of the three collections. This severely constrains the Bengali-Hindi evaluation. Only idioms for which a Hindi entry with an English figurative meaning existed could be paired, limiting the number of bn-hi pairs to 4,096. Many Bengali idioms with clear Hindi equivalents may have been left unmatched,

simply because the Hindi resource is incomplete. The small Hindi pool also renders the retrieval task for bn-hi, artificially easier in one sense (fewer distractors) but harder in another (reduced statistical power to detect differences between models). The small size also likely contributes to LaBSE’s unexpectedly low bn-hi similarity scores; with only 200 Hindi embeddings in the space, the model’s representation of Hindi idiomatic semantics may be less robust, especially if its training corpus under-represents Hindi figurative language.

6.2.5 Near-Zero Triplet Cohesion

Triplet cohesion, the metric that most directly signals true three-way cross-lingual alignment, is vanishingly small for both models and is zero for LaBSE. Even mSBERT’s 1.15% figure, while statistically non-zero, is low in absolute terms. This indicates that perfect multi-way clustering of idioms by figurative meaning remains an unsolved challenge. Several factors conspire against high triplet cohesion: the dataset imbalance, the noise in Hindi alignment, the fixed number of clusters ($k=50$) that may not correspond to the natural number of semantic categories, and the inherent difficulty of forcing three separate linguistic instantiations of a figurative concept into exactly one cluster. The low cohesion values place an important boundary on the study’s positive claims: models can align idioms in a pairwise sense and can generate a similarity signal clearly above chance, but they are not yet reliably organising the full multilingual figurative lexicon into coherent conceptual groupings.

6.2.6 Model-Specific Weaknesses

Despite the promising overall results, each model exhibited distinct limitations that constrain its practical utility for cross-lingual idiom processing. A recurring issue across multiple models was the trade-off between semantic similarity confidence and discriminative ranking power: models that produced high mean cosine scores for equivalent pairs (such as mSBERT) often struggled to separate true matches from near-neighbours in retrieval tasks, resulting in poor precision-at- k metrics. Conversely, models that excelled at retrieval (LaBSE) showed lower similarity confidence and, in some cases, counter-intuitive patterns such as lower similarity scores for linguistically closer language pairs. The most severe failure was observed in a general-purpose multilingual encoder (XLM-R), which suffered from complete embedding collapse for Bengali and Hindi, producing near-uniform vectors that made any distinction between idioms impossible. This collapse rendered the model useless for both similarity scoring and retrieval, despite superficially high cosine values. Meanwhile, the monolingual Bengali model (BanglaBERT), while strong within its single language, was entirely incapable of cross-lingual transfer, limiting its applicability to the core research problem.

Another class of weaknesses related to cluster coherence and language mixing. Models that achieved high cluster purity, meaning that most clusters contained idioms from only one language, failed at triplet cohesion, the most stringent test of cross-lingual alignment. Conversely, the model with the best triplet cohesion still produced a positive but low absolute rate (under 2%), indicating that even the strongest model cannot reliably co-locate equivalent idioms from three languages in the same cluster. Furthermore, all models showed degraded

performance on the Bengali-Hindi pair relative to what linguistic proximity would predict, except for one model that actually performed better on that pair but still underperformed in retrieval. These patterns suggest that no single model is universally superior; the choice of model must be guided by the specific task (similarity estimation, retrieval, or clustering) and the language pair in question. The embedding space geometry required for each task appears to be partially at odds, implying that future work may need to develop task-specific fine-tuning or ensemble approaches.

6.2.7 Evaluation Constraints

The evaluation setup, while comprehensive, has several constraints. First, all idioms were evaluated as isolated phrases rather than embedded in full sentences. In natural usage, context disambiguates idiomatic meaning and provides additional semantic cues; a model might perform better or worse when idioms appear within sentences. The current results therefore characterise decontextualised idiomatic alignment, which is a necessary first step but not the final word on real-world utility. Second, the clustering analysis relied on a single clustering algorithm (K-Means) with a predetermined k of 50. Different clustering methods (e.g., agglomerative clustering, DBSCAN) or different values of k could yield different structural insights. The choice of k was a practical compromise, but the observed purity and cohesion are partly artefacts of this choice. Third, the study did not include a direct comparison against translation-based baselines, such as translating the Bengali idiom to English using a commercial MT system and then applying a monolingual similarity model. The claim of “without translation” is a feature of the models themselves, but the absence of a translation-based control limits the ability to quantify how much better or worse direct embedding matching is relative to a pipeline that does use translation.

6.3 Conclusion

This chapter has detailed the principal drawbacks of the present study. They include the unavoidable dataset imbalance, imperfect ground truth for bn-hi pairs, the off-the-shelf nature of the models, the limited Hindi resource, low absolute triplet cohesion, model-specific weaknesses in retrieval and similarity, and methodological constraints such as isolated phrase evaluation and fixed clustering parameters. None of these drawbacks invalidate the core findings: the models demonstrably exceed random baselines, exhibit complementary strengths, and reveal linguistically meaningful patterns. However, the drawbacks do impose caution on overgeneralising the results and highlight clear pathways for improvement. Recognising these limitations transparently is essential for framing the future work discussed in the next chapter and for ensuring that the dissertation’s contributions are interpreted with appropriate scholarly nuance.

CHAPTER 07: CONCLUSION

7.1 Recapitulation of Research Problem and Objectives

This dissertation set out to investigate a deceptively simple but profoundly challenging question: can state-of-the-art multilingual sentence embedding models identify semantically equivalent idioms across Bengali, English, and Hindi without relying on any translation mechanism whatsoever? Idioms, defined by their non-compositional semantics, constitute a frontier for natural language understanding; they resist literal interpretation and demand that a model apprehend figurative meaning that transcends lexical forms. The difficulty is compounded when the languages involved belong to different typological branches and when one of them, Bengali, suffers from a severe scarcity of digital language resources.

The primary research objective was to compare **four** models: **mSBERT** (paraphrase-distilled student), **LaBSE** (dual-encoder trained on translation pairs), **XLM-RoBERTa-base** (general-purpose multilingual encoder), and **BanglaBERT** (monolingual Bengali baseline). The evaluation assessed their capacity to perform cross-lingual idiom matching across three tasks (semantic similarity scoring, cross-lingual retrieval, and cross-lingual clustering), supplemented by monolingual Bengali analysis. A curated cross-lingual idiom triplet dataset was constructed. The entire investigation was conducted under the strict constraint of no translation, testing the genuine cross-lingual semantic knowledge encoded in the models' vector spaces.

7.2 Summary of Key Findings

The empirical results, derived from a dataset of 9,061 Bengali-English pairs, 4,096 Bengali-Hindi pairs, and a clustering corpus of 23,544 embeddings, yield a multifaceted picture of the models' capabilities and limitations.

7.2.1 Semantic Similarity Scoring

Both mSBERT and LaBSE assign significantly positive cosine similarity to equivalent idiom pairs, well above the random baseline of near zero. mSBERT emerges as the stronger performer, with mean cosine of 0.438 for Bengali-English and a notably high 0.563 for Bengali-Hindi. LaBSE produces lower and more variable scores (bn-en: 0.388; bn-hi: 0.288). **XLM-R** produces near-perfect mean similarity (≈ 0.993) for both pairs, but this is a symptom of **catastrophic embedding collapse**; all vectors are nearly identical, rendering the scores meaningless. BanglaBERT is evaluated only monolingually, achieving a mean bn-bn similarity of 0.831. mSBERT's elevated bn-hi score reflects linguistic proximity (both Indo-Aryan languages), a pattern absent in LaBSE and erased in XLM-R.

The differences were statistically significant. mSBERT’s scores reflect a robust general confidence in semantic equivalence and, intriguingly, an amplification for the Bengali-Hindi pair. This amplification aligns with the linguistic reality that Bengali and Hindi, as co-members of the Indo-Aryan family, share etymological, morphological, and cultural-conceptual substrates that mSBERT’s paraphrase-oriented training appears to capture. LaBSE’s weaker and more variable similarity scores, particularly for bn-hi, suggest that its translation-pair-based objective, while excellent for alignment, results in a more conservative and less calibrated confidence scale for figurative expressions, especially for language pairs underrepresented in its training data.

7.2.2 Cross-Lingual Retrieval

Retrieval, the most practically relevant task, sees LaBSE as the clear winner across MRR, P@1, P@5, and Hit@10 for both language pairs. Although absolute scores are modest due to extreme difficulty (most queries have no correct match, large candidate pools), all models exceed random baselines, except XLM-R which performs near-random for bn-en. LaBSE’s training with additive margin softmax provides fine-grained ranking ability. mSBERT lags behind but remains above baseline. XLM-R’s collapsed space makes it useless for retrieval (P@1 = 0 for bn-en). Retrieval metrics are substantially higher for bn-hi than bn-en for all functioning models, again underscoring the role of linguistic proximity.

7.2.3 Cross-Lingual Clustering

Clustering (K=50, all languages) reveals divergent embedding geometries. **LaBSE** achieves the most language-agnostic space: positive silhouette (0.041), lowest purity (0.745, indicating cross-lingual mixing), and the highest triplet cohesion (0.018, i.e., 1.8% of Bengali-English-Hindi triplets perfectly co-clustered). **mSBERT** has negative silhouette (-0.018), high purity (0.902, language-segregated), and lower but non-zero triplet cohesion (0.011). **XLM-R** has positive silhouette (0.042) and extremely high purity (0.985) but **zero triplet cohesion**; it separates languages completely, and the positive silhouette is an artefact of collapse. **BanglaBERT** is not evaluated cross-lingually.

7.2.4 Monolingual Bengali Evaluation

As a diagnostic, monolingual tasks show that **LaBSE** spreads Bengali idioms widely (low mean similarity 0.331) and achieves the only positive silhouette (0.034) for Bengali-only clustering. **BanglaBERT** and **mSBERT** produce more compact representations (similarities 0.831 and 0.681, respectively) but negative silhouettes. **XLM-R** collapses even monolingually (similarity 0.994, silhouette -0.051). Thus, LaBSE is best for fine-grained discrimination; BanglaBERT is strong for monolingual applications.

7.2.5 Overall Model Comparison

The final verdict across all cross-lingual tasks (excluding XLM-R’s invalid wins) gives **LaBSE the overall victory**, with advantages in retrieval, triplet cohesion, cluster mixing, and monolingual silhouette. mSBERT wins on similarity-confidence and reflecting linguistic proximity. XLM-R is disqualified due to embedding collapse. BanglaBERT is the best choice for purely monolingual Bengali tasks. The complementary nature of mSBERT and LaBSE suggests task-dependent selection: LaBSE for search-oriented applications, mSBERT for measuring figurative similarity or exploratory semantic mapping.

7.3 Contributions of the Study

This dissertation makes several contributions to cross-lingual NLP and figurative language processing.

1. **A new cross-lingual idiom dataset** aligned across Bengali, English, and Hindi, constructed using transparent alignment strategies (human-curated and keyword-overlap). It fills a resource gap for low-resource figurative language evaluation.
2. **The first systematic multi-task comparison of four models** (mSBERT, LaBSE, XLM-R, BanglaBERT) on non-compositional, figurative semantics. Prior work focused on literal sentence alignment; this thesis extends evaluation to idioms.
3. **Empirical evidence on embedding collapse** in general-purpose multilingual encoders (XLM-R) when applied to fine-grained tasks with low-resource languages, serving as a cautionary finding for future research.
4. **Demonstration of linguistic proximity effects** within the Indo-Aryan family: mSBERT’s higher performance on Bengali-Hindi vs. Bengali-English confirms that multilingual models can leverage shared etymological and cultural features.
5. **A rigorous, reproducible evaluation framework** combining similarity scores, retrieval metrics, clustering analysis (including triplet cohesion), and monolingual diagnostics, extensible to other language pairs and figurative constructs.

7.4 Limitations

The findings must be considered in light of the limitations detailed in the preceding chapter. The dataset suffers from a marked imbalance in idiom counts across languages, with Bengali heavily overrepresented. The ground truth for Bengali-Hindi pairs, generated through automatic keyword overlap, contains an estimated 22% of erroneous alignments, adding noise to the evaluation. Both models were used off-the-shelf without fine-tuning, so the results represent a lower bound on achievable performance. The Hindi dataset is small, limiting the breadth and statistical power of the bn-hi analysis. Triplet cohesion remains extremely low, indicating that perfect three-way alignment is still an unsolved challenge. Finally, idioms were evaluated in isolation, and the clustering analysis was performed with a fixed K-Means

parameterisation. These limitations, while substantial, do not undermine the core conclusions but delineate the boundaries within which they apply and point to clear directions for improvement.

7.5 Final Remarks

This dissertation has demonstrated that the figurative heart of language - its idioms - is not entirely opaque to multilingual neural models. Without any translation, without any explicit rules, and without any task-specific fine-tuning, pretrained sentence embeddings can place the Bengali “আকাশ কুসুম,” the English “build castles in the air,” and the Hindi “हवाई किले बनाना” in meaningful proximity within a shared mathematical space. The fact that this happens at all, given the linguistic and cultural distances involved, is a testament to modern representation learning. However, the picture is nuanced. LaBSE emerges as the most balanced model, excelling at retrieval and cluster mixing. mSBERT captures linguistic proximity and similarity confidence. XLM-R collapses entirely, serving as a cautionary example. BanglaBERT reminds us that monolingual strength does not guarantee cross-lingual transfer.

The low absolute retrieval scores and the still-minuscule triplet cohesion reveal how far the field remains from robust, reliable cross-lingual figurative understanding. The journey from “measurable” to “usable” will require richer parallel idiom data, careful fine-tuning, and perhaps architectural innovations that explicitly model figurative language. This thesis, by charting the current capabilities of four diverse models on a challenging new benchmark, hopes to contribute a useful map for that ongoing journey.

CHAPTER 08: FUTURE SCOPE

8.1 Introduction

The research presented in this dissertation has established a baseline for cross-lingual idiom matching using off-the-shelf multilingual sentence embedding models, and it has simultaneously illuminated a number of limitations and open questions. The measured performance levels, while clearly above random chance, remain modest in absolute terms, particularly for retrieval and three-way clustering. This gap between the demonstrated potential and the requirements of practical deployment defines a rich agenda for future investigation. In this chapter, we outline the most promising directions for extending the current work, organised around data enrichment, model adaptation, linguistic expansion, contextualised evaluation, methodological refinement, and integration into real-world applications. These proposals are grounded in the empirical findings and acknowledged drawbacks of the present study and are intended to guide subsequent research toward robust, usable cross-lingual figurative language technology.

8.2 Data Enrichment

The most immediate and impactful improvement lies in strengthening the dataset that underpins the evaluation.

8.2.1 Expansion of the Hindi Idiom Collection

The current Hindi dataset of approximately 200 idioms is severely undersized relative to the Bengali and English collections. A concentrated effort to compile or construct a Hindi idiom inventory of at least 1,000 entries, each annotated with its figurative meaning in English, would dramatically improve the breadth and statistical reliability of the Bengali-Hindi evaluation. Such a resource could be built through a combination of scraping online Hindi idiom repositories, collaborating with native speaker linguists, and translating existing English idiom meaning descriptions. A larger and more representative Hindi pool would reduce the noise in the keyword-overlap alignment, permit a more accurate estimation of retrieval metrics across a wider range of queries, and give the clustering algorithm a fairer view of Hindi’s distribution in the multilingual embedding space.

8.2.2 Human Verification of Cross-Lingual Triplets

The ground truth for the triplet dataset currently contains an estimated 22% error rate for Bengali-Hindi pairs due to the automated alignment process. A systematic human annotation campaign, even on a modest scale of 200 to 300 randomly selected triplets, would produce a

high-quality gold standard subset. This subset could serve two purposes: as a clean evaluation set against which models can be compared without the confounding factor of label noise, and as a fine-tuning seed for domain adaptation. Bilingual speakers of Bengali-Hindi, Bengali-English, and ideally trilingual speakers, could be recruited to judge whether two idioms convey the same figurative meaning, using a Likert scale or binary decision protocol. The Inter-Annotator Agreement (IAA) would quantify the inherent difficulty of the task and provide a human performance ceiling.

8.2.3 Addressing Dataset Imbalance

The current dataset’s severe skew toward Bengali (8,800 entries) distorts some of the clustering metrics, particularly purity. A balanced or less imbalanced dataset, in which English and Hindi idiom counts are increased to the order of thousands, would allow for a cleaner analysis of cross-lingual mixing in the embedding space. Future work could also explore sampling strategies that equalise language representation without discarding the richness of the larger Bengali collection, such as stratified sampling during clustering or weighted evaluation metrics that account for class sizes.

8.3 Model Adaptation and Fine-Tuning

The present study deliberately froze the pretrained models to test their out-of-the-box capabilities. However, the results strongly imply that domain adaptation could yield substantial gains.

8.3.1 Fine-Tuning on Verified Idiom Pairs

Using a gold standard set of a few hundred verified Bengali-English and Bengali-Hindi pairs, both models could be fine-tuned with a contrastive loss or a triplet loss. This would explicitly teach the model to pull equivalent idioms closer while pushing non-equivalent idioms apart in the shared space. Even a few epochs of domain-specific fine-tuning are likely to sharpen the similarity distributions, reduce the variance, and, most critically, improve retrieval precision. A comparison of fine-tuned mSBERT and LaBSE would reveal which architecture benefits more from idiom-specific supervision, a question with both theoretical and practical implications.

8.3.2 Few-Shot and Cross-Lingual Transfer

Given the scarcity of annotated data for low-resource languages, few-shot learning paradigms are particularly relevant. A model fine-tuned on Bengali-English idiom pairs could be tested on its zero-shot transfer to Bengali-Hindi matching, probing whether the figurative alignment learned with English as a bridge generalises to a language pair that was not seen during fine-tuning. Similarly, meta-learning or prompt-based approaches could be explored to rapidly adapt models to new languages with minimal annotated data.

8.3.3 Idiom-Aware Pretraining

On a more ambitious horizon, one could envisage incorporating idiom resources into the pretraining phase of multilingual models. This might involve creating a supplementary pretraining corpus where idiomatic expressions are marked, or constructing synthetic parallel data that emphasises figurative meaning equivalence, for example by pairing an idiom with its literal paraphrase in multiple languages. Such pretraining would bake idiomatic awareness into the foundation model, potentially benefiting a wide range of downstream figurative language tasks.

8.4 Extension to Additional Languages and Language Families

The three-language setup of this dissertation can serve as a template for a larger cross-linguistic study. Immediate candidates for inclusion are other South Asian languages, such as Urdu (which shares substantial lexical and cultural overlap with both Hindi and Bengali), Odia, Assamese, and Tamil. Tamil, a Dravidian language, would provide a contrasting case: it is unrelated to the Indo-Aryan family and would test the models’ ability to generalise figurative alignment across deep typological divides. Expanding the study to include European languages like Spanish, German, or Russian would further probe the universality of the observed findings and the models’ sensitivity to shared cultural metaphor structures. Such a multilingual expansion would require careful construction of aligned idiom datasets for each new language, but the methodology is directly transferable.

8.5 Contextualised Idiom Evaluation

The current work evaluated idioms as isolated phrases. In real texts, idioms appear embedded in sentences that provide disambiguating context. Two idioms that are semantically equivalent in isolation might diverge when placed in different syntactic or pragmatic environments, or conversely, a model that fails to recognise an isolated idiom equivalence might succeed when contextual cues are present. Future research should construct a sentence-level evaluation dataset where each idiom is placed in a natural, representative sentence in its language, and the cross-lingual matching is performed between these sentence-embedded idioms. This would test the models’ ability to integrate contextual information with stored figurative knowledge and would provide a more ecologically valid measure of practical utility. Both monolingual and cross-lingual sentence similarity benchmarks could be adapted to include idiomatic test cases.

8.6 Integration with Downstream Applications

Beyond intrinsic evaluation, the cross-lingual idiom matching capability can be tested within real-world NLP pipelines.

8.6.1 Machine Translation

Idiom matching models could be integrated as a post-processing module in neural machine translation systems. When the MT system encounters a source idiom, a cross-lingual idiom retriever could propose an idiomatic equivalent in the target language, either replacing a literal translation or providing alternatives to the user. Evaluating the impact of such a module on translation quality, particularly on the preservation of figurative meaning, would be a significant applied contribution.

8.6.2 Language Learning and Lexicography

A cross-lingual idiom search tool, possibly powered by LaBSE for retrieval and mSBERT for similarity ranking, could assist language learners in discovering equivalent idiomatic expressions in their target language. It could also support lexicographers in compiling multilingual idiom dictionaries by suggesting candidate alignments that can then be verified by human experts. A prototype system with a simple user interface could be built and tested with actual learners to gauge perceived utility and accuracy.

8.6.3 Cultural Analytics and Digital Humanities

Idiom equivalence patterns can serve as windows into shared and divergent cultural conceptualisations. Clustering embeddings of idioms from many languages could reveal universal metaphor themes and culture-specific variations. Future work could apply the embedding and clustering methodology to large cross-cultural idiom corpora, yielding quantitative insights for disciplines such as comparative literature, anthropology, and cognitive linguistics.

8.7 Methodological Refinements

Several technical improvements to the evaluation framework are worth pursuing.

8.7.1 Clustering Methodology

The current reliance on K-Means with a fixed k of 50 is a simplification. Future studies should experiment with different clustering algorithms (agglomerative clustering, DBSCAN, spectral clustering), optimise the number of clusters using quantitative criteria (elbow method, gap statistic), and introduce soft clustering or density-based approaches that can better capture overlapping semantic categories. Evaluation metrics such as adjusted mutual information or homogeneity and completeness scores could supplement purity and silhouette to give a more nuanced picture of cross-lingual cluster quality.

8.7.2 Translation-Based Baselines

While the core contribution of this work is the demonstration of translation-free matching, future research should include strong translation-based baselines for comparison. For instance, translating the Bengali idiom to English using a commercial or state-of-the-art MT system, then applying a monolingual SBERT similarity model, would quantify the performance delta. Such a comparison would clarify whether direct multilingual embeddings are competitive with, or even superior to, pipelines that include translation, especially for low-resource language pairs where MT quality may be poor.

8.7.3 Statistical Rigor

Expanding the number of runs with different random seeds for clustering, applying bootstrap confidence intervals to metric estimates, and conducting more extensive significance testing across all evaluation dimensions would enhance the robustness of the conclusions. The present study reported key statistical tests for the similarity scoring task; extending this rigour to retrieval and clustering, perhaps via permutation tests or Bayesian approaches, would be valuable.

8.8 Conclusion

The future work agenda emerging from this dissertation is broad and interdisciplinary. It spans the practicalities of data curation for low-resource languages, the technicalities of fine-tuning and model adaptation, the linguistic ambition of expanding to new language families, and the integrative goal of embedding this technology into applications that serve human communication and cultural understanding. The present study’s demonstration that cross-lingual idiomatic alignment is measurable with current off-the-shelf models provides both a foundation and a motivation to pursue these extensions. As multilingual models continue to grow in capacity and coverage, and as the NLP community pays increasing attention to figurative and non-compositional language, the methods and findings laid out in this dissertation can serve as a stepping stone toward systems that truly understand the richness and subtlety of human idiomatic expression across the world’s languages.

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